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JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

# Sequences and Series

JEE Main Mathematics · Previous Year Questions with Answer Key

| 341 | 193 | 2002 – 2026 | 111 | English |
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## MCQ (Single Correct Answer) 181 questions

**Q1.** JEE Main 2026 (Online) 8th April Evening Shift

Let $\alpha = 3 + 4 + 8 + 9 + 13 + 14 + \ldots$ upto 40 terms. If $(\tan\beta)^{\frac{\alpha}{1020}}$ is a root of the equation $x^2 + x - 2 = 0, \beta \in \left(0, \frac{\pi}{2}\right)$, then $\sin^2\beta + 3\cos^2\beta$ is equal to :

(A) 2 (B) $\frac{7}{4}$ (C) $\frac{5}{2}$ (D) $\frac{3}{2}$

Open solution

**Q2.** JEE Main 2026 (Online) 6th April Evening Shift

Consider the quadratic equation $\left(n^2 - 2n + 2\right)x^2 - 3x + \left(n^2 - 2n + 2\right)^2 = 0, n \in \mathbf{R}$. Let $\alpha$ be the minimum value of the product of its roots and $\beta$ be the maximum value of the sum of its roots. Then the sum of the first six terms of the G.P., whose first term is $\alpha$ and the common ratio is $\frac{\alpha}{\beta}$, is :

(A) $\frac{61}{37}$ (B) $\frac{121}{81}$ (C) $\frac{364}{243}$ (D) $\frac{1093}{729}$

Open solution

**Q3.** JEE Main 2026 (Online) 6th April Evening Shift

The sum $1 + \frac{1}{2}\left(1^2 + 2^2\right) + \frac{1}{3}\left(1^2 + 2^2 + 3^2\right) + \ldots$ upto 10 terms is equal to :

(A) 130 (B) 155 (C) $\frac{315}{2}$ (D) $\frac{325}{2}$

Open solution

**Q4.** JEE Main 2026 (Online) 6th April Morning Shift

The value of $1^3 - 2^3 + 3^3 - \ldots + 15^3$ is:

(A) 1706 (B) 1856 (C) 1982 (D) 2403

Open solution

**Q5.** JEE Main 2026 (Online) 6th April Morning Shift

The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8 . If the first term of the A.P. is equal to the common ratio of the G.P. and the first term of the G.P. is equal to common difference of the A.P., then the sum of all possible values of the first term of the G.P. is:

(A) $\frac{34}{9}$ (B) $\frac{34}{13}$ (C) $\frac{32}{9}$ (D) $\frac{32}{13}$

Open solution

**Q6.** JEE Main 2026 (Online) 5th April Evening Shift

Let $\alpha, \beta$ be the roots of the equation $x^2 - x + \mathrm{p} = 0$ and $\gamma, \delta$ be the roots the equation $x^2 - 4x + \mathrm{q} = 0; p, q \in \mathbf{Z}$. If $\alpha, \beta, \gamma, \delta$ are in G.P., then $|p + q|$ equals :

(A) 16 (B) 32 (C) 34 (D) 38

Open solution

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**Q7.** JEE Main 2026 (Online) 5th April Evening Shift

If the sum of the first 10 terms of the series $\frac{1}{1+1^4\times4}+\frac{2}{1+2^4\times4}+\frac{3}{1+3^4\times4}+\frac{4}{1+4^4\times4}+\ldots\ldots$ is $\frac{m}{n}, \gcd(m,n)=1$, then $m+n$ is equal to :

(A) 256 (B) 264 (C) 276 (D) 284

Open solution

**Q8.** JEE Main 2026 (Online) 5th April Evening Shift

Let $A_1, A_2, A_3, \ldots\ldots\ldots, A_{39}$ be 39 arithmetic means between the numbers 59 and 159. Then the mean of $A_{25}, A_{28}, A_{31}$ and $A_{36}$ is equal to :

(A) 129 (B) 136 (C) 131.50 (D) 134

Open solution

**Q9.** JEE Main 2026 (Online) 5th April Morning Shift

Let the sum of the first $n$ terms of an A.P. be $3n^2+5n$. Then the sum of squares of the first 10 terms of the A.P. is:

(A) 10220 (B) 12860 (C) 15220 (D) 19780

Open solution

**Q10.** JEE Main 2026 (Online) 5th April Morning Shift

$\sum_{n=1}^{10}\left(\frac{528}{n(n+1)(n+2)}\right)$ is equal to:

(A) 65 (B) 130 (C) 220 (D) 440

Open solution

**Q11.** JEE Main 2026 (Online) 4th April Morning Shift

The first term of an A.P. of 30 non-negative terms is $\frac{10}{3}$. If the sum of this A.P. is the cube of its last term, then its common difference is:

(A) $\frac{5}{87}$ (B) $\frac{25}{83}$ (C) $\frac{15}{29}$ (D) $\frac{5}{29}$

Open solution

**Q12.** JEE Main 2026 (Online) 2nd April Evening Shift

Let $a_1, a_2, a_3, \ldots$ be an A.P. and $g_1=a_1, g_2, g_3, \ldots$ be an increasing G.P. If $a_1=a_2+g_2=1$ and $a_3+g_3=4$, then $a_{10}+g_5$ is equal to:

(A) 81 (B) 76 (C) 62 (D) 55

Open solution

**Q13.** JEE Main 2026 (Online) 2nd April Evening Shift

The sum $\frac{1^3}{1}+\frac{1^3+2^3}{1+3}+\frac{1^3+2^3+3^3}{1+3+5}+\ldots$ up to 8 terms, is :

(A) 70 (B) 71 (C) 72 (D) 73

Open solution

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**Q14.** JEE Main 2026 (Online) 2nd April Morning Shift

Let A be the set of first 101 terms of an A.P., whose first term is 1 and the common difference is 5 and let B be the set of first 71 terms of an A.P., whose first term is 9 and the common difference is 7. Then the number of elements in $A \cap B$, which are divisible by 3, is :

Open solution

(A) 4 (B) 5 (C) 6 (D) 7

**Q15.** JEE Main 2026 (Online) 28th January Evening Shift

Let the arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}$ be $\frac{5}{16}$, $a > 2$. If $\alpha$ is such that $a$, 4, $\alpha$, $b$ are in A.P., then the equation $\alpha x^2 - ax + 2(\alpha - 2b) = 0$ has :

Open solution

(A) one root in $(1, 4)$ and another in $(-2, 0)$ (B) one root in $(0, 2)$ and another in $(-4, -2)$

(C) both roots in the interval $(-2, 0)$ (D) complex roots of magnitude less than 2

**Q16.** JEE Main 2026 (Online) 28th January Evening Shift

$\frac{6}{3^{26}} + \frac{10 \cdot 1}{3^{25}} + \frac{10 \cdot 2}{3^{24}} + \frac{10 \cdot 2^2}{3^{23}} + \ldots + \frac{10 \cdot 2^{24}}{3}$ is equal to :

Open solution

(A) $2^{26}$ (B) $3^{25}$ (C) $3^{26}$ (D) $2^{25}$

**Q17.** JEE Main 2026 (Online) 28th January Morning Shift

The value of $\sum_{k=1}^{\infty} (-1)^{k+1} \left(\frac{k(k+1)}{k!}\right)$ is

Open solution

(A) e/2 (B) $\sqrt{e}$ (C) 2/e (D) 1/e

**Q18.** JEE Main 2026 (Online) 28th January Morning Shift

The common difference of the A.P.: $a_1, a_2, \ldots, a_m$ is 13 more than the common difference of the A.P.: $b_1, b_2, \ldots, b_n$. If $b_{31} = -277, b_{43} = -385$ and $a_{78} = 327$, then $a_1$ is equal to

Open solution

(A) 21 (B) 19 (C) 24 (D) 16

**Q19.** JEE Main 2026 (Online) 24th January Evening Shift

Let $a_1, a_2, a_3, a_4$ be an A.P. of four terms such that each term of the A.P. and its common difference $l$ are integers. If $a_1 + a_2 + a_3 + a_4 = 48$ and $a_1 a_2 a_3 a_4 + l^4 = 361$, then the largest term of the A.P. is equal to

Open solution

(A) 27 (B) 24 (C) 23 (D) 21

**Q20.** JEE Main 2026 (Online) 24th January Evening Shift

$\left(\frac{1}{3} + \frac{4}{7}\right) + \left(\frac{1}{3^2} + \frac{1}{3} \times \frac{4}{7} + \frac{4^2}{7^2}\right) + \left(\frac{1}{3^3} + \frac{1}{3^2} \times \frac{4}{7} + \frac{1}{3} \times \frac{4^2}{7^2} + \frac{4^3}{7^3}\right) + \ldots$ upto infinite terms, is equal to

Open solution

(A) $\frac{7}{4}$ (B) $\frac{4}{3}$ (C) $\frac{6}{5}$ (D) $\frac{5}{2}$

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**Q21.** JEE Main 2026 (Online) 24th January Morning Shift

Let $729, 81, 9, 1, \ldots$ be a sequence and $\mathrm{P}_n$ denote the product of the first $n$ terms of this sequence.

If $2\sum_{n=1}^{40}(\mathrm{P}_n)^{\frac{1}{n}} = \frac{3^{\alpha}-1}{3^{\beta}}$ and $\gcd(\alpha,\beta)=1$, then

$\alpha+\beta$ is equal to

(A) 73 (B) 74 (C) 75 (D) 76

Open solution

**Q22.** JEE Main 2026 (Online) 24th January Morning Shift

Consider an A.P.: $a_1, a_2, \ldots, a_n; a_1 > 0$. If $a_2 - a_1 = \frac{-3}{4}$, $a_n = \frac{1}{4}a_1$, and $\sum_{i=1}^{n} a_i = \frac{525}{2}$, then $\sum_{i=1}^{17} a_i$ is equal to

(A) 238 (B) 136 (C) 476 (D) 952

Open solution

**Q23.** JEE Main 2026 (Online) 23rd January Evening Shift

Let $\sum_{k=1}^{n} a_k = \alpha n^2 + \beta n$. If $a_{10} = 59$ and $a_6 = 7a_1$, then $\alpha+\beta$ is equal to :

(A) 3 (B) 5 (C) 7 (D) 12

Open solution

**Q24.** JEE Main 2026 (Online) 22nd January Morning Shift

If the sum of the first four terms of an A.P. is 6 and the sum of its first six terms is 4 , then the sum of its first twelve terms is

(A) -26 (B) -20 (C) -24 (D) -22

Open solution

**Q25.** JEE Main 2026 (Online) 21st January Evening Shift

The positive integer **n**, for which the solutions of the equation

$x(x+2)+(x+2)(x+4)+\cdots+(x+2n-2)(x+2n)=\frac{8n}{3}$ are two consecutive even integers, is :

(A) 3 (B) 6 (C) 9 (D) 12

Open solution

**Q26.** JEE Main 2026 (Online) 21st January Evening Shift

Let $a_1, \frac{a_2}{2}, \frac{a_3}{2^2}, \ldots, \frac{a_{10}}{2^9}$ be a G.P. of common ratio $\frac{1}{\sqrt{2}}$. If $a_1+a_2+\ldots+a_{10}=62$, then $a_1$ is equal to:

(A) $\sqrt{2}-1$ (B) $2(\sqrt{2}-1)$ (C) $2-\sqrt{2}$ (D) $2(2-\sqrt{2})$

Open solution

**Q27.** JEE Main 2026 (Online) 21st January Morning Shift

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms such that $a_2 \cdot a_3 \cdot a_4 = 64$ and $a_1+a_3+a_5=\frac{813}{7}$. Then $a_3+a_5+a_7$ is equal to :

(A) 3256 (B) 3252 (C) 3248 (D) 3244

Open solution

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**Q28.** JEE Main 2025 (Online) 8th April Evening Shift

If $\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+\ldots\infty=\frac{\pi^4}{90}$,

$\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\ldots\infty=\alpha$,

$\frac{1}{2^4}+\frac{1}{4^4}+\frac{1}{6^4}+\ldots\infty=\beta$,

then $\frac{\alpha}{\beta}$ is equal to :

(A) 23 (B) 14 (C) 18 (D) 15

Open solution

**Q29.** JEE Main 2025 (Online) 7th April Evening Shift

Let $a_n$ be the $n^{th}$ term of an A.P. If $S_n=a_1+a_2+a_3+\ldots+a_n=700$, $a_6=7$ and $S_7=7$, then $a_n$ is equal to :

(A) 65 (B) 56 (C) 70 (D) 64

Open solution

**Q30.** JEE Main 2025 (Online) 7th April Evening Shift

If the sum of the second, fourth and sixth terms of a G.P. of positive terms is 21 and the sum of its eighth, tenth and twelfth terms is 15309, then the sum of its first nine terms is :

(A) 757 (B) 755 (C) 750 (D) 760

Open solution

**Q31.** JEE Main 2025 (Online) 7th April Morning Shift

Let $x_1, x_2, x_3, x_4$ be in a geometric progression. If $2, 7, 9, 5$ are subtracted respectively from $x_1, x_2, x_3, x_4$, then the resulting numbers are in an arithmetic progression. Then the value of $\frac{1}{24}(x_1x_2x_3x_4)$ is:

(A) 18 (B) 216 (C) 36 (D) 72

Open solution

**Q32.** JEE Main 2025 (Online) 4th April Evening Shift

If the sum of the first 20 terms of the series $\frac{4\cdot1}{4+3\cdot1^2+1^4}+\frac{4\cdot2}{4+3\cdot2^2+2^4}+\frac{4\cdot3}{4+3\cdot3^2+3^4}+\frac{4\cdot4}{4+3\cdot4^2+4^4}+\ldots$ is $\frac{\mathrm{m}}{\mathrm{n}}$, where m and n are coprime, then $\mathrm{m}+\mathrm{n}$ is equal to :

(A) 423 (B) 421 (C) 422 (D) 420

Open solution

**Q33.** JEE Main 2025 (Online) 4th April Evening Shift

Consider two sets A and B, each containing three numbers in A.P. Let the sum and the product of the elements of A be 36 and p respectively and the sum and the product of the elements of B be 36 and $q$ respectively. Let d and D be the common differences of $\mathrm{AP}'\mathrm{s}$ in $A$ and $B$ respectively such that $D=d+3, d>0$. If $\frac{p+q}{p-q}=\frac{19}{5}$, then $\mathrm{p}-\mathrm{q}$ is equal to

(A) 540 (B) 450 (C) 600 (D) 630

Open solution

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**Q34.** JEE Main 2025 (Online) 4th April Morning Shift

Let $A = \{1, 6, 11, 16, \ldots\}$ and $B = \{9, 16, 23, 30, \ldots\}$ be the sets consisting of the first 2025 terms of two arithmetic progressions. Then $n(A \cup B)$ is

(A) 3814 (B) 4003 (C) 4027 (D) 3761

Open solution

**Q35.** JEE Main 2025 (Online) 4th April Morning Shift

$1 + 3 + 5^2 + 7 + 9^2 + \ldots$ upto 40 terms is equal to

(A) 40870 (B) 41880 (C) 43890 (D) 33980

Open solution

**Q36.** JEE Main 2025 (Online) 3rd April Evening Shift

The sum $1 + \frac{1+3}{2!} + \frac{1+3+5}{3!} + \frac{1+3+5+7}{4!} + \ldots$ upto $\infty$ terms, is equal to

(A) $3e$ (B) $2e$ (C) $4e$ (D) $6e$

Open solution

**Q37.** JEE Main 2025 (Online) 3rd April Morning Shift

Let $a_1, a_2, a_3, \ldots$. be a G.P. of increasing positive numbers. If $a_3 a_5 = 729$ and $a_2 + a_4 = \frac{111}{4}$, then $24\,(a_1 + a_2 + a_3)$ is equal to

(A) 128 (B) 129 (C) 131 (D) 130

Open solution

**Q38.** JEE Main 2025 (Online) 3rd April Morning Shift

The sum $1 + 3 + 11 + 25 + 45 + 71 + \ldots$ upto 20 terms, is equal to

(A) 7240 (B) 8124 (C) 7130 (D) 6982

Open solution

**Q39.** JEE Main 2025 (Online) 2nd April Evening Shift

The number of terms of an A.P. is even; the sum of all the odd terms is 24 , the sum of all the even terms is 30 and the last term exceeds the first by $\frac{21}{2}$. Then the number of terms which are integers in the A.P. is :

(A) 6 (B) 4 (C) 8 (D) 10

Open solution

**Q40.** JEE Main 2025 (Online) 2nd April Morning Shift

Let $a_1, a_2, a_3, \ldots$ be in an A.P. such that $\sum_{k=1}^{12} a_{2k-1} = -\frac{72}{5} a_1, a_1 \neq 0$. If $\sum_{k=1}^{n} a_k = 0$, then $n$ is :

(A) 18 (B) 17 (C) 11 (D) 10

Open solution

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**Q41.** JEE Main 2025 (Online) 29th January Morning Shift

Consider an A. P. of positive integers, whose sum of the first three terms is 54 and the sum of the first twenty terms lies between 1600 and 1800. Then its $11^{th}$ term is :

(A) 108 (B) 90 (C) 122 (D) 84

Open solution

**Q42.** JEE Main 2025 (Online) 28th January Evening Shift

For positive integers $n$, if $4a_n = \left(n^2 + 5n + 6\right)$ and $S_n = \sum_{k=1}^{n} \left(\frac{1}{a_k}\right)$, then the value of $507S_{2025}$ is :

(A) 540 (B) 675 (C) 1350 (D) 135

Open solution

**Q43.** JEE Main 2025 (Online) 28th January Morning Shift

Let $\langle a_n \rangle$ be a sequence such that $a_0 = 0, a_1 = \frac{1}{2}$ and $2a_{n+2} = 5a_{n+1} - 3a_n, n = 0, 1, 2, 3, \ldots$. Then $\sum_{k=1}^{100} a_k$ is equal to

(A) $3a_{100} + 100$ (B) $3a_{100} - 100$ (C) $3a_{99} - 100$ (D) $3a_{99} + 100$

Open solution

**Q44.** JEE Main 2025 (Online) 28th January Morning Shift

Let $T_r$ be the $r^{th}$ term of an A.P. If for some $m$, $T_m = \frac{1}{25}$, $T_{25} = \frac{1}{20}$, and $20\sum_{r=1}^{25} T_r = 13$, then $5m\sum_{r=m}^{2m} T_r$ is equal to

(A) 98 (B) 126 (C) 112 (D) 142

Open solution

**Q45.** JEE Main 2025 (Online) 24th January Evening Shift

In an arithmetic progression, if $S_{40} = 1030$ and $S_{12} = 57$, then $S_{30} - S_{10}$ is equal to :

(A) 525 (B) 505 (C) 510 (D) 515

Open solution

**Q46.** JEE Main 2025 (Online) 24th January Evening Shift

If $7 = 5 + \frac{1}{7}(5 + \alpha) + \frac{1}{7^2}(5 + 2\alpha) + \frac{1}{7^3}(5 + 3\alpha) + \ldots\ldots\ldots\ldots\infty$, then the value of $\alpha$ is :

(A) $\frac{1}{7}$ (B) 1 (C) $\frac{6}{7}$ (D) 6

Open solution

**Q47.** JEE Main 2025 (Online) 24th January Morning Shift

Let $S_n = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \ldots$ upto $n$ terms. If the sum of the first six terms of an A.P. with first term -p and common difference p is $\sqrt{2026\, S_{2025}}$, then the absolute difference between $20^{th}$ and $15^{th}$ terms of the A.P. is

(A) 20 (B) 45 (C) 90 (D) 25

Open solution

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**Q48.** JEE Main 2025 (Online) 23rd January Morning Shift

If the first term of an A.P. is 3 and the sum of its first four terms is equal to one-fifth of the sum of the next four terms, then the sum of the first 20 terms is equal to

(A) $-120$ (B) $-1200$ (C) $-1080$ (D) $-1020$

Open solution

**Q49.** JEE Main 2025 (Online) 22nd January Evening Shift

Suppose that the number of terms in an A.P. is $2k, k \in N$. If the sum of all odd terms of the A.P. is 40 , the sum of all even terms is 55 and the last term of the A.P. exceeds the first term by 27 , then k is equal to:

(A) 8 (B) 6 (C) 4 (D) 5

Open solution

**Q50.** JEE Main 2025 (Online) 22nd January Morning Shift

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms. If $a_1 a_5 = 28$ and $a_2 + a_4 = 29$, then $a_6$ is equal to:

(A) 812 (B) 784 (C) 628 (D) 526

Open solution

**Q51.** JEE Main 2024 (Online) 9th April Evening Shift

Let $a, ar, ar^2, \ldots\ldots\ldots$ be an infinite G.P. If $\sum_{n=0}^{\infty} ar^n = 57$ and $\sum_{n=0}^{\infty} a^3 r^{3n} = 9747$, then $a + 18r$ is equal to

(A) 27 (B) 38 (C) 31 (D) 46

Open solution

**Q52.** JEE Main 2024 (Online) 9th April Morning Shift

If the sum of the series $\frac{1}{1\cdot(1+\mathrm{d})} + \frac{1}{(1+\mathrm{d})(1+2\,\mathrm{d})} + \ldots + \frac{1}{(1+9\,\mathrm{d})(1+10\,\mathrm{d})}$ is equal to 5, then $50\,\mathrm{d}$ is equal to :

(A) 5 (B) 10 (C) 15 (D) 20

Open solution

**Q53.** JEE Main 2024 (Online) 8th April Evening Shift

In an increasing geometric progression of positive terms, the sum of the second and sixth terms is $\frac{70}{3}$ and the product of the third and fifth terms is 49. Then the sum of the $4^{\text{th}}$, $6^{\text{th}}$ and $8^{\text{th}}$ terms is equal to:

(A) 78 (B) 96 (C) 91 (D) 84

Open solution

**Q54.** JEE Main 2024 (Online) 6th April Evening Shift

Let $ABC$ be an equilateral triangle. A new triangle is formed by joining the middle points of all sides of the triangle $ABC$ and the same process is repeated infinitely many times. If $\mathrm{P}$ is the sum of perimeters and $Q$ is the sum of areas of all the triangles formed in this process, then :

(A) $\mathrm{P}^2 = 72\sqrt{3}\mathrm{Q}$ (B) $\mathrm{P}^2 = 36\sqrt{3}\mathrm{Q}$

(C) $\mathrm{P} = 36\sqrt{3}\mathrm{Q}^2$ (D) $\mathrm{P}^2 = 6\sqrt{3}\mathrm{Q}$

Open solution

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**Q55.** JEE Main 2024 (Online) 6th April Evening Shift

A software company sets up m number of computer systems to finish an assignment in 17 days. If 4 computer systems crashed on the start of the second day, 4 more computer systems crashed on the start of the third day and so on, then it took 8 more days to finish the assignment. The value of $\mathrm{m}$ is equal to:

(A) 125 (B) 160 (C) 150 (D) 180

Open solution

**Q56.** JEE Main 2024 (Online) 5th April Evening Shift

For $x \geqslant 0$, the least value of $\mathrm{K}$, for which $4^{1+x}+4^{1-x}, \frac{\mathrm{K}}{2}, 16^{x}+16^{-x}$ are three consecutive terms of an A.P., is equal to :

(A) 10 (B) 4 (C) 8 (D) 16

Open solution

**Q57.** JEE Main 2024 (Online) 5th April Morning Shift

If $\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\ldots+\frac{1}{\sqrt{99}+\sqrt{100}}=m$ and $\frac{1}{1 \cdot 2}+\frac{1}{2 \cdot 3}+\ldots+\frac{1}{99 \cdot 100}=\mathrm{n}$, then the point $(\mathrm{m}, \mathrm{n})$ lies on the line

(A) $11(x-1)-100y=0$ (B) $11x-100y=0$

(C) $11(x-1)-100(y-2)=0$ (D) $11(x-2)-100(y-1)=0$

Open solution

**Q58.** JEE Main 2024 (Online) 4th April Evening Shift

The value of $\frac{1 \times 2^{2}+2 \times 3^{2}+\ldots+100 \times(101)^{2}}{1^{2} \times 2+2^{2} \times 3+\ldots+100^{2} \times 101}$ is

(A) $\frac{305}{301}$ (B) $\frac{306}{305}$ (C) $\frac{32}{31}$ (D) $\frac{31}{30}$

Open solution

**Q59.** JEE Main 2024 (Online) 4th April Evening Shift

Let three real numbers $a, b, c$ be in arithmetic progression and $a+1, b, c+3$ be in geometric progression. If $a>10$ and the arithmetic mean of $a, b$ and $c$ is 8, then the cube of the geometric mean of $a, b$ and $c$ is

(A) 120 (B) 316 (C) 312 (D) 128

Open solution

**Q60.** JEE Main 2024 (Online) 4th April Morning Shift

Let the first three terms 2, p and q, with $q \neq 2$, of a G.P. be respectively the $7^{\text{th}}, 8^{\text{th}}$ and $13^{\text{th}}$ terms of an A.P. If the $5^{\text{th}}$ term of the G.P. is the $n^{\text{th}}$ term of the A.P., then $n$ is equal to:

(A) 151 (B) 177 (C) 163 (D) 169

Open solution

**Q61.** JEE Main 2024 (Online) 1st February Evening Shift

Let $S_{n}$ denote the sum of the first $n$ terms of an arithmetic progression. If $S_{10}=390$ and the ratio of the tenth and the fifth terms is $15:7$, then $\mathrm{S}_{15}-\mathrm{S}_{5}$ is equal to :

(A) 800 (B) 890 (C) 790 (D) 690

Open solution

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**Q62.** JEE Main 2024 (Online) 1st February Morning Shift

Let $3, a, b, c$ be in A.P. and $3, a-1, b+1, c+9$ be in G.P. Then, the arithmetic mean of $a, b$ and $c$ is :

(A) -4 (B) -1 (C) 13 (D) 11

Open solution

**Q63.** JEE Main 2024 (Online) 31st January Evening Shift

Let $2^{nd}, 8^{th}$ and $44^{th}$ terms of a non-constant A. P. be respectively the $1^{st}, 2^{nd}$ and $3^{rd}$ terms of a G. P. If the first term of the A. P. is 1, then the sum of its first 20 terms is equal to -

(A) 990 (B) 980 (C) 960 (D) 970

Open solution

**Q64.** JEE Main 2024 (Online) 31st January Morning Shift

For $0 < c < b < a$, let $(a+b-2c)x^2 + (b+c-2a)x + (c+a-2b) = 0$ and $\alpha \neq 1$ be one of its roots. Then, among the two statements

(I) If $\alpha \in (-1, 0)$, then $b$ cannot be the geometric mean of $a$ and $c$

(II) If $\alpha \in (0, 1)$, then $b$ may be the geometric mean of $a$ and $c$

(A) only (II) is true (B) Both (I) and (II) are true

(C) only (I) is true (D) Neither (I) nor (II) is true

Open solution

**Q65.** JEE Main 2024 (Online) 31st January Morning Shift

The sum of the series $\frac{1}{1-3\cdot1^2+1^4} + \frac{2}{1-3\cdot2^2+2^4} + \frac{3}{1-3\cdot3^2+3^4} + \ldots$ up to 10 -terms is

(A) $\frac{45}{109}$ (B) $-\frac{55}{109}$ (C) $\frac{55}{109}$ (D) $-\frac{45}{109}$

Open solution

**Q66.** JEE Main 2024 (Online) 30th January Evening Shift

Let $a$ and $b$ be two distinct positive real numbers. Let $11^{th}$ term of a GP, whose first term is $a$ and third term is $b$, is equal to $p^{th}$ term of another GP, whose first term is $a$ and fifth term is $b$. Then $p$ is equal to

(A) 20 (B) 24 (C) 21 (D) 25

Open solution

**Q67.** JEE Main 2024 (Online) 30th January Morning Shift

Let $S_n$ denote the sum of first $n$ terms of an arithmetic progression. If $S_{20} = 790$ and $S_{10} = 145$, then $S_{15} - S_5$ is :

(A) 405 (B) 390 (C) 410 (D) 395

Open solution

**Q68.** JEE Main 2024 (Online) 29th January Evening Shift

If $\log_e a, \log_e b, \log_e c$ are in an A.P. and $\log_e a - \log_e 2b, \log_e 2b - \log_e 3c, \log_e 3c - \log_e a$ are also in an A.P, then $a : b : c$ is equal to

(A) $6 : 3 : 2$ (B) $9 : 6 : 4$ (C) $25 : 10 : 4$ (D) $16 : 4 : 1$

Open solution

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**Q69.** JEE Main 2024 (Online) 29th January Evening Shift

If each term of a geometric progression $a_1, a_2, a_3, \ldots$ with $a_1 = \frac{1}{8}$ and $a_2 \neq a_1$, is the arithmetic mean of the next two terms and $S_n = a_1 + a_2 + \ldots\ldots + a_n$, then $S_{20} - S_{18}$ is equal to

(A) $-2^{15}$ (B) $2^{15}$ (C) $-2^{18}$ (D) $2^{18}$

Open solution

**Q70.** JEE Main 2024 (Online) 29th January Morning Shift

If in a G.P. of 64 terms, the sum of all the terms is 7 times the sum of the odd terms of the G.P, then the common ratio of the G.P. is equal to

(A) 7 (B) 6 (C) 5 (D) 4

Open solution

**Q71.** JEE Main 2024 (Online) 29th January Morning Shift

In an A.P., the sixth term $a_6 = 2$. If the product $a_1 a_4 a_5$ is the greatest, then the common difference of the A.P. is equal to

(A) $\frac{2}{3}$ (B) $\frac{5}{8}$ (C) $\frac{3}{2}$ (D) $\frac{8}{5}$

Open solution

**Q72.** JEE Main 2024 (Online) 27th January Evening Shift

The $20^{\text{th}}$ term from the end of the progression $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \ldots, -129\frac{1}{4}$ is :

(A) $-115$ (B) $-100$ (C) $-110$ (D) $-118$

Open solution

**Q73.** JEE Main 2024 (Online) 27th January Morning Shift

The number of common terms in the progressions

$4, 9, 14, 19, \ldots\ldots$, up to $25^{\text{th}}$ term and

$3, 6, 9, 12, \ldots\ldots$, up to $37^{\text{th}}$ term is :

(A) 9 (B) 8 (C) 5 (D) 7

Open solution

**Q74.** JEE Main 2023 (Online) 15th April Morning Shift

Let $A_1$ and $A_2$ be two arithmetic means and $G_1, G_2, G_3$ be three geometric means of two distinct positive numbers. Then $G_1^4 + G_2^4 + G_3^4 + G_1^2 G_3^2$ is equal to :

(A) $(A_1 + A_2)^2 G_1 G_3$ (B) $(A_1 + A_2) G_1^2 G_3^2$

(C) $2(A_1 + A_2) G_1^2 G_3^2$ (D) $2(A_1 + A_2) G_1 G_3$

Open solution

**Q75.** JEE Main 2023 (Online) 13th April Evening Shift

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive numbers. Let the sum of its $6^{\text{th}}$ and $8^{\text{th}}$ terms be 2 and the product of its $3^{\text{rd}}$ and $5^{\text{th}}$ terms be $\frac{1}{9}$. Then $6(a_2 + a_4)(a_4 + a_6)$ is equal to

(A) $2\sqrt{2}$ (B) 2 (C) $3\sqrt{3}$ (D) 3

Open solution

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**Q76.** JEE Main 2023 (Online) 13th April Morning Shift

Let $s_1, s_2, s_3, \ldots, s_{10}$ respectively be the sum to 12 terms of 10 A.P. s whose first terms are $1, 2, 3, \ldots .10$ and the common differences are $1, 3, 5, \ldots \ldots, 19$ respectively. Then $\sum_{i=1}^{10} s_i$ is equal to :

(A) 7360 (B) 7220 (C) 7260 (D) 7380

Open solution

**Q77.** JEE Main 2023 (Online) 11th April Evening Shift

Let $a, b, c$ and $d$ be positive real numbers such that $a+b+c+d=11$. If the maximum value of $a^5b^3c^2d$ is $3750\beta$, then the value of $\beta$ is

(A) 110 (B) 108 (C) 90 (D) 55

Open solution

**Q78.** JEE Main 2023 (Online) 11th April Morning Shift

Let $x_1, x_2, \ldots, x_{100}$ be in an arithmetic progression, with $x_1 = 2$ and their mean equal to 200 . If $y_i = i\,(x_i - i), 1 \leq i \leq 100$, then the mean of $y_1, y_2, \ldots, y_{100}$ is :

(A) 10051.50 (B) 10049.50 (C) 10100 (D) 10101.50

Open solution

**Q79.** JEE Main 2023 (Online) 10th April Morning Shift

Let the first term $\alpha$ and the common ratio r of a geometric progression be positive integers. If the sum of squares of its first three terms is 33033, then the sum of these three terms is equal to

(A) 241 (B) 231 (C) 220 (D) 210

Open solution

**Q80.** JEE Main 2023 (Online) 31st January Evening Shift

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $a_7 = 3$, the product $a_1a_4$ is minimum and the sum of its first $n$ terms is zero, then $n! - 4a_{n(n+2)}$ is equal to :

(A) 24 (B) $\frac{381}{4}$ (C) 9 (D) $\frac{33}{4}$

Open solution

**Q81.** JEE Main 2023 (Online) 31st January Morning Shift

If the sum and product of four positive consecutive terms of a G.P., are 126 and 1296 , respectively, then the sum of common ratios of all such GPs is

(A) 7 (B) 14 (C) 3 (D) $\frac{9}{2}$

Open solution

**Q82.** JEE Main 2023 (Online) 30th January Evening Shift

Let $a, b, c > 1, a^3, b^3$ and $c^3$ be in A.P., and $\log_a b, \log_c a$ and $\log_b c$ be in G.P. If the sum of first 20 terms of an A.P., whose first term is $\frac{a+4b+c}{3}$ and the common difference is $\frac{a-8b+c}{10}$ is $-444$, then $abc$ is equal to :

(A) 343 (B) 216 (C) $\frac{343}{8}$ (D) $\frac{125}{8}$

Open solution

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**Q83.** JEE Main 2023 (Online) 24th January Morning Shift

For three positive integers p, q, r, $x^{pq^2} = y^{qr} = z^{p^2r}$ and r = pq + 1 such that 3, 3 $\log_y x$, 3 $\log_z y$, 7 $\log_x z$ are in A.P. with common difference $\frac{1}{2}$. Then r-p-q is equal to

(A) 12 (B) $-6$ (C) 6 (D) 2

Open solution

**Q84.** JEE Main 2022 (Online) 30th July Evening Shift

For $|\alpha| \geq 1$, let $5^{4-2\alpha}$, 63, $5^{2\alpha-1}$ be in A.P. with the common difference $d$. Let $S_n$ denote the sum of the first $n$ terms of an A.P., whose first term is $\alpha^2$ and the common difference is $d$. If $S_{30} - S_{15} = 30k$, then $k$ is equal to

(A) 715 (B) 695 (C) 684 (D) 683

Open solution

**Q85.** JEE Main 2022 (Online) 27th July Evening Shift

Let the sum of an infinite G.P., whose first term is a and the common ratio is r, be 5 . Let the sum of its first five terms be $\frac{98}{25}$. Then the sum of the first 21 terms of an AP, whose first term is $10ar$, $n^{th}$ term is $a_n$ and the common difference is $10ar^2$, is equal to :

(A) $21\, a_{11}$ (B) $22\, a_{11}$ (C) $15\, a_{16}$ (D) $14\, a_{16}$

Open solution

**Q86.** JEE Main 2022 (Online) 27th July Morning Shift

Suppose $a_1, a_2, \ldots, a_n, \ldots$ be an arithmetic progression of natural numbers. If the ratio of the sum of first five terms to the sum of first nine terms of the progression is $5 : 17$ and , $110 < a_{15} < 120$, then the sum of the first ten terms of the progression is equal to

(A) 290 (B) 380 (C) 460 (D) 510

Open solution

**Q87.** JEE Main 2022 (Online) 26th July Morning Shift

Consider two G.Ps. 2, $2^2$, $2^3$, ..... and 4, $4^2$, $4^3$, .... of 60 and n terms respectively. If the geometric mean of all the 60 + n terms is $(2)^{\frac{225}{8}}$, then $\sum_{k=1}^{n} k(n-k)$ is equal to :

(A) 560 (B) 1540 (C) 1330 (D) 2600

Open solution

**Q88.** JEE Main 2022 (Online) 28th June Evening Shift

If n arithmetic means are inserted between a and 100 such that the ratio of the first mean to the last mean is 1 : 7 and a + n = 33, then the value of n is :

(A) 21 (B) 22 (C) 23 (D) 24

Open solution

**Q89.** JEE Main 2022 (Online) 28th June Morning Shift

Let $A_1, A_2, A_3, \ldots\ldots$ be an increasing geometric progression of positive real numbers. If $A_1A_3A_5A_7 = \frac{1}{1296}$ and $A_2 + A_4 = \frac{7}{36}$, then the value of $A_6 + A_8 + A_{10}$ is equal to

(A) 33 (B) 37 (C) 43 (D) 47

Open solution

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**Q90.** JEE Main 2022 (Online) 27th June Evening Shift

If $a_1, a_2, a_3$ ..... and $b_1, b_2, b_3$ ..... are A.P., and $a_1 = 2$, $a_{10} = 3$, $a_1b_1 = 1 = a_{10}b_{10}$, then $a_4\, b_4$ is equal to -

(A) $\frac{35}{27}$ (B) 1 (C) $\frac{27}{28}$ (D) $\frac{28}{27}$

Open solution

**Q91.** JEE Main 2022 (Online) 27th June Morning Shift

$x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$, where a, b, c are in A.P. and |a| < 1, |b| < 1, |c| < 1, abc $\neq$ 0, then :

(A) x, y, z are in A.P.

(B) x, y, z are in G.P.

(C) $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P.

(D) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1 - (a + b + c)$

Open solution

**Q92.** JEE Main 2022 (Online) 25th June Morning Shift

If $\frac{1}{2 \cdot 3^{10}} + \frac{1}{2^2 \cdot 3^9} + \ldots\ldots + \frac{1}{2^{10} \cdot 3} = \frac{K}{2^{10} \cdot 3^{10}}$, then the remainder when K is divided by 6 is :

(A) 1 (B) 2 (C) 3 (D) 5

Open solution

**Q93.** JEE Main 2022 (Online) 24th June Evening Shift

Let x, y > 0. If $x^3y^2 = 2^{15}$, then the least value of 3x + 2y is

(A) 30 (B) 32 (C) 36 (D) 40

Open solution

**Q94.** JEE Main 2022 (Online) 24th June Morning Shift

If $\{a_i\}_{i=1}^{n}$, where n is an even integer, is an arithmetic progression with common difference 1, and $\sum_{i=1}^{n} a_i = 192$, $\sum_{i=1}^{n/2} a_{2i} = 120$, then n is equal to :

(A) 48 (B) 96 (C) 92 (D) 104

Open solution

**Q95.** JEE Main 2021 (Online) 1st September Evening Shift

Let $a_1, a_2, \ldots\ldots\ldots, a_{21}$ be an AP such that $\sum_{n=1}^{20} \frac{1}{a_n a_{n+1}} = \frac{4}{9}$. If the sum of this AP is 189, then $a_6a_{16}$ is equal to :

(A) 57 (B) 72 (C) 48 (D) 36

Open solution

**Q96.** JEE Main 2021 (Online) 31st August Evening Shift

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $\frac{a_1+a_2+\ldots+a_{10}}{a_1+a_2+\ldots+a_p} = \frac{100}{p^2}$, p $\neq$ 10, then $\frac{a_{11}}{a_{10}}$ is equal to :

(A) $\frac{19}{21}$ (B) $\frac{100}{121}$ (C) $\frac{21}{19}$ (D) $\frac{121}{100}$

Open solution

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**Q97.** JEE Main 2021 (Online) 31st August Morning Shift

Three numbers are in an increasing geometric progression with common ratio r. If the middle number is doubled, then the new numbers are in an arithmetic progression with common difference d. If the fourth term of GP is 3 $r^2$, then $r^2 - d$ is equal to :

(A) $7 - 7\sqrt{3}$ (B) $7 + \sqrt{3}$ (C) $7 - \sqrt{3}$ (D) $7 + 3\sqrt{3}$

Open solution

**Q98.** JEE Main 2021 (Online) 26th August Morning Shift

If the sum of an infinite GP a, ar, $ar^2$, $ar^3$, ..... is 15 and the sum of the squares of its each term is 150, then the sum of $ar^2$, $ar^4$, $ar^6$, ..... is :

(A) $\frac{5}{2}$ (B) $\frac{1}{2}$ (C) $\frac{25}{2}$ (D) $\frac{9}{2}$

Open solution

**Q99.** JEE Main 2021 (Online) 25th July Morning Shift

Let $S_n$ be the sum of the first n terms of an arithmetic progression. If $S_{3n} = 3S_{2n}$, then the value of $\frac{S_{4n}}{S_{2n}}$ is :

(A) 6 (B) 4 (C) 2 (D) 8

Open solution

**Q100.** JEE Main 2021 (Online) 22th July Evening Shift

Let $S_n$ denote the sum of first n-terms of an arithmetic progression. If $S_{10} = 530$, $S_5 = 140$, then $S_{20} - S_6$ is equal to:

(A) 1862 (B) 1842 (C) 1852 (D) 1872

Open solution

**Q101.** JEE Main 2021 (Online) 18th March Evening Shift

Let $S_1$ be the sum of first 2n terms of an arithmetic progression. Let $S_2$ be the sum of first 4n terms of the same arithmetic progression. If $(S_2 - S_1)$ is 1000, then the sum of the first 6n terms of the arithmetic progression is equal to :

(A) 7000 (B) 1000 (C) 3000 (D) 5000

Open solution

**Q102.** JEE Main 2021 (Online) 26th February Morning Shift

In an increasing geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of $4^{th}$, $6^{th}$ and $8^{th}$ terms is equal to :

(A) 30 (B) 32 (C) 26 (D) 35

Open solution

**Q103.** JEE Main 2021 (Online) 25th February Evening Shift

The minimum value of $f(x) = a^{a^x} + a^{1-a^x}$, where a, $x \in R$ and a > 0, is equal to :

(A) $a + \frac{1}{a}$ (B) 2a (C) a + 1 (D) $2\sqrt{a}$

Open solution

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**Q104.** JEE Main 2020 (Online) 6th September Evening Slot

The common difference of the A.P. $b_1, b_2, \ldots, b_m$ is 2 more than the common difference of A.P. $a_1, a_2, \ldots, a_n$. If $a_{40} = -159$, $a_{100} = -399$ and $b_{100} = a_{70}$, then $b_1$ is equal to :

(A) 127 (B) 81 (C) −127 (D) -81

Open solution

**Q105.** JEE Main 2020 (Online) 6th September Morning Slot

Let a , b, c , d and p be any non zero distinct real numbers such that $(a^2 + b^2 + c^2)p^2 - 2(ab + bc + cd)p + (b^2 + c^2 + d^2) = 0$. Then :

(A) a, c, p are in G.P. (B) a, b, c, d are in G.P.

(C) a, b, c, d are in A.P. (D) a, c, p are in A.P.

Open solution

**Q106.** JEE Main 2020 (Online) 5th September Evening Slot

If the sum of the second, third and fourth terms of a positive term G.P. is 3 and the sum of its sixth, seventh and eighth terms is 243, then the sum of the first 50 terms of this G.P. is :

(A) $\frac{2}{13}\left(3^{50}-1\right)$ (B) $\frac{1}{13}\left(3^{50}-1\right)$

(C) $\frac{1}{26}\left(3^{49}-1\right)$ (D) $\frac{1}{26}\left(3^{50}-1\right)$

Open solution

**Q107.** JEE Main 2020 (Online) 5th September Morning Slot

If $3^{2\sin 2\alpha-1}$, 14 and $3^{4-2\sin 2\alpha}$ are the first three terms of an A.P. for some $\alpha$, then the sixth term of this A.P. is:

(A) 66 (B) 81 (C) 65 (D) 78

Open solution

**Q108.** JEE Main 2020 (Online) 4th September Evening Slot

The minimum value of $2^{\sin x} + 2^{\cos x}$ is :

(A) $2^{-1+\sqrt{2}}$ (B) $2^{1-\frac{1}{\sqrt{2}}}$ (C) $2^{1-\sqrt{2}}$ (D) $2^{-1+\frac{1}{\sqrt{2}}}$

Open solution

**Q109.** JEE Main 2020 (Online) 4th September Evening Slot

Let $a_1, a_2, \ldots$, an be a given A.P. whose common difference is an integer and $S_n = a_1 + a_2 + \ldots + a_n$. If $a_1 = 1$, $a_n = 300$ and $15 \leq n \leq 50$, then the ordered pair $(S_{n-4}, a_{n-4})$ is equal to:

(A) (2480, 249) (B) (2480, 248) (C) (2490, 248) (D) (2490, 249)

Open solution

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**Q110.** JEE Main 2020 (Online) 3rd September Morning Slot

If the first term of an A.P. is 3 and the sum of its first 25 terms is equal to the sum of its next 15 terms, then the common difference of this A.P. is :

(A) $\frac{1}{4}$ (B) $\frac{1}{5}$ (C) $\frac{1}{7}$ (D) $\frac{1}{6}$

Open solution

**Q111.** JEE Main 2020 (Online) 2nd September Evening Slot

If the sum of first 11 terms of an A.P.,
$a_1, a_2, a_3, \ldots$ is 0 ($a \neq 0$), then the sum of the A.P.,
$a_1, a_3, a_5, \ldots, a_{23}$ is $ka_1$, where k is equal to :

(A) $\frac{121}{10}$ (B) $-\frac{121}{10}$ (C) $\frac{72}{5}$ (D) $-\frac{72}{5}$

Open solution

**Q112.** JEE Main 2020 (Online) 2nd September Morning Slot

The sum of the first three terms of a G.P. is S and their product is 27. Then all such S lie in :

(A) $[-3, \infty)$ (B) $(-\infty, 9]$
(C) $(-\infty, -9] \cup [-3, \infty)$ (D) $(-\infty, -3] \cup [9, \infty)$

Open solution

**Q113.** JEE Main 2020 (Online) 9th January Evening Slot

Let $a_n$ be the $n^{th}$ term of a G.P. of positive terms.

$$\sum_{n=1}^{100} a_{2n+1} = 200 \text{ and } \sum_{n=1}^{100} a_{2n} = 100,$$

then $\sum_{n=1}^{200} a_n$ is equal to :

(A) 150 (B) 175 (C) 225 (D) 300

Open solution

**Q114.** JEE Main 2020 (Online) 8th January Evening Slot

If the $10^{th}$ term of an A.P. is $\frac{1}{20}$ and its $20^{th}$ term is $\frac{1}{10}$, then the sum of its first 200 terms is

(A) 100 (B) $100\frac{1}{2}$ (C) $50\frac{1}{4}$ (D) 50

Open solution

**Q115.** JEE Main 2020 (Online) 8th January Morning Slot

Let $f : \mathbf{R} \to \mathbf{R}$ be such that for all $x \in R$
$(2^{1+x} + 2^{1-x})$, $f(x)$ and $(3^x + 3^{-x})$ are in A.P.,
then the minimum value of $f(x)$ is

(A) 2 (B) 0 (C) 3 (D) 4

Open solution

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**Q116.** JEE Main 2020 (Online) 7th January Evening Slot

Let $a_1$ , $a_2$ , $a_3$ ,..... be a G.P. such that
$a_1$ < 0, $a_1$ + $a_2$ = 4 and $a_3$ + $a_4$ = 16.
If $\sum_{i=1}^{9} a_i = 4\lambda$, then $\lambda$ is equal to:

(A) 171 (B) -171 (C) -513 (D) $\frac{511}{3}$

Open solution

**Q117.** JEE Main 2020 (Online) 7th January Morning Slot

Five numbers are in A.P. whose sum is 25 and product is 2520. If one of these five numbers is -$\frac{1}{2}$ , then the greatest number amongst them is:

(A) $\frac{21}{2}$ (B) 27 (C) 7 (D) 16

Open solution

**Q118.** JEE Main 2019 (Online) 12th April Evening Slot

If $a_1$, $a_2$, $a_3$, ..... are in A.P. such that $a_1 + a_7 + a_{16} = 40$, then the sum of the first 15 terms of this A.P. is :

(A) 120 (B) 200 (C) 150 (D) 280

Open solution

**Q119.** JEE Main 2019 (Online) 12th April Morning Slot

If $\alpha$ and $\beta$ are the roots of the equation $375x^2 - 25x - 2 = 0$, then $\lim_{n\to\infty} \sum_{r=1}^{n} \alpha^r + \lim_{n\to\infty} \sum_{r=1}^{n} \beta^r$ is equal to :

(A) $\frac{7}{116}$ (B) $\frac{29}{358}$ (C) $\frac{1}{12}$ (D) $\frac{21}{346}$

Open solution

**Q120.** JEE Main 2019 (Online) 12th April Morning Slot

Let $S_n$ denote the sum of the first n terms of an A.P. If $S_4 = 16$ and $S_6 = -48$, then $S_{10}$ is equal to :

(A) - 320 (B) - 380 (C) - 460 (D) - 210

Open solution

**Q121.** JEE Main 2019 (Online) 10th April Evening Slot

Let $a$, b and c be in G.P. with common ratio r, where $a \neq 0$ and $0 < r \leq \frac{1}{2}$ . If $3a$, 7b and 15c are the first three terms of an A.P., then the $4^{th}$ term of this A.P. is :

(A) $a$ (B) $\frac{7}{3}a$ (C) $5a$ (D) $\frac{2}{3}a$

Open solution

**Q122.** JEE Main 2019 (Online) 10th April Evening Slot

Let $a_1$, $a_2$, $a_3$,.....be an A.P. with $a_6 = 2$. Then the common difference of this A.P., which maximises the product $a_1a_4a_5$, is :

(A) $\frac{3}{2}$ (B) $\frac{6}{5}$ (C) $\frac{8}{5}$ (D) $\frac{2}{3}$

Open solution

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**Q123.** JEE Main 2019 (Online) 10th April Morning Slot

If $a_1, a_2, a_3, \ldots\ldots\ldots\ldots a_n$ are in A.P. and $a_1 + a_4 + a_7 + \ldots\ldots\ldots + a_{16} = 114$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to :

(A) 38 (B) 98 (C) 76 (D) 64

Open solution

**Q124.** JEE Main 2019 (Online) 9th April Evening Slot

If the sum and product of the first three terms in an A.P. are 33 and 1155, respectively, then a value of its $11^{th}$ term is :-

(A) −25 (B) −36 (C) 25 (D) −35

Open solution

**Q125.** JEE Main 2019 (Online) 9th April Morning Slot

Let the sum of the first n terms of a non-constant A.P., $a_1, a_2, a_3, \ldots.$ be $50n + \frac{n(n-7)}{2}A$, where A is a constant. If d is the common difference of this A.P., then the ordered pair (d, $a_{50}$) is equal to

(A) (A, 50+45A) (B) (50, 50+45A) (C) (A, 50+46A) (D) (50, 50+46A)

Open solution

**Q126.** JEE Main 2019 (Online) 8th April Evening Slot

If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct?

(A) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P. (B) d, e, f are in A.P

(C) d, e, f are in G.P (D) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

Open solution

**Q127.** JEE Main 2019 (Online) 8th April Morning Slot

The sum of all natural numbers 'n' such that 100 < n < 200 and H.C.F. (91, n) > 1 is :

(A) 3221 (B) 3121 (C) 3203 (D) 3303

Open solution

**Q128.** JEE Main 2019 (Online) 12th January Evening Slot

If $\sin^4\alpha + 4\cos^4\beta + 2 = 4\sqrt{2}\sin\alpha\cos\beta;\ \alpha, \beta \in [0, \pi]$,
then $\cos(\alpha + \beta) - \cos(\alpha - \beta)$ is equal to :

(A) $-\sqrt{2}$ (B) 0 (C) $-1$ (D) $\sqrt{2}$

Open solution

**Q129.** JEE Main 2019 (Online) 12th January Evening Slot

If ${}^nC_4, {}^nC_5$ and ${}^nC_6$ are in A.P., then n can be :

(A) 11 (B) 12 (C) 9 (D) 14

Open solution

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**Q130.** JEE Main 2019 (Online) 12th January Morning Slot

The product of three consecutive terms of a G.P. is 512. If 4 is added to each of the first and the second of these terms, the three terms now form an A.P. Then the sum of the original three terms of the given G.P. is :

(A) 36 (B) 28 (C) 32 (D) 24

Open solution

**Q131.** JEE Main 2019 (Online) 11th January Evening Slot

If $19^{th}$ term of a non-zero A.P. is zero, then its ($49^{th}$ term) : ($29^{th}$ term) is :

(A) 2 : 1 (B) 4 : 1 (C) 1 : 3 (D) 3 : 1

Open solution

**Q132.** JEE Main 2019 (Online) 11th January Evening Slot

Let x, y be positive real numbers and m, n positive integers. The maximum value of the expression $\frac{x^m y^n}{(1+x^{2m})(1+y^{2n})}$ is :

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{m+n}{6mn}$ (D) 1

Open solution

**Q133.** JEE Main 2019 (Online) 11th January Morning Slot

The sum of an infinite geometric series with positive terms is 3 and the sum of the cubes of its terms is $\frac{27}{19}$.Then the common ratio of this series is :

(A) $\frac{4}{9}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{2}{9}$

Open solution

**Q134.** JEE Main 2019 (Online) 11th January Morning Slot

Let $a_1, a_2, \ldots\ldots, a_{10}$ be a G.P. If $\frac{a_3}{a_1} = 25$, then $\frac{a_9}{a_5}$ equals

(A) $5^3$ (B) $2(5^2)$ (C) $4(5^2)$ (D) $5^4$

Open solution

**Q135.** JEE Main 2019 (Online) 10th January Evening Slot

Let $a_1, a_2, a_3, \ldots\ldots a_{10}$ be in G.P. with $a_i > 0$ for i = 1, 2, ....., 10 and S be the set of pairs (r, k), r, k $\in$ N (the set of natural numbers) for which

$$\begin{vmatrix} \log_e a_1{}^r a_2{}^k & \log_e a_2{}^r a_3{}^k & \log_e a_3{}^r a_4{}^k \\ \log_e a_4{}^r a_5{}^k & \log_e a_5{}^r a_6{}^k & \log_e a_6{}^r a_7{}^k \\ \log_e a_7{}^r a_8{}^k & \log_e a_8{}^r a_9{}^k & \log_e a_9{}^r a_{10}{}^k \end{vmatrix} = 0.$$

Then the number of elements in S, is -

(A) 10 (B) 4 (C) 2 (D) infinitely many

Open solution

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**Q136.** JEE Main 2019 (Online) 10th January Morning Slot

The sum of all two digit positive numbers which when divided by 7 yield 2 or 5 as remainder is -

(A) 1356 (B) 1256 (C) 1365 (D) 1465

Open solution

**Q137.** JEE Main 2019 (Online) 9th January Evening Slot

Let a, b and c be the $7^{th}$, $11^{th}$ and $13^{th}$ terms respectively of a non-constant A.P. If these are also three consecutive terms of a G.P., then $\frac{a}{c}$ is equal to :

(A) 2 (B) $\frac{1}{2}$ (C) $\frac{7}{13}$ (D) 4

Open solution

**Q138.** JEE Main 2019 (Online) 9th January Morning Slot

If a, b, c be three distinct real numbers in G.P. and a + b + c = xb , then x **cannot** be

(A) 2 (B) -3 (C) 4 (D) -2

Open solution

**Q139.** JEE Main 2019 (Online) 9th January Morning Slot

Let $a_1, a_2, \ldots\ldots, a_{30}$ be an A.P.,

$S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{(2i-1)}$.

If $a_5$ = 27 and S - 2T = 75, then $a_{10}$ is equal to :

(A) 47 (B) 42 (C) 52 (D) 57

Open solution

**Q140.** JEE Main 2018 (Online) 16th April Morning Slot

Let $\frac{1}{x_1}, \frac{1}{x_2}, \ldots, \frac{1}{x_n}$ ($x_i \neq 0$ for i = 1, 2, ..., n) be in A.P. such that $x_1$=4 and $x_{21}$ = 20. If n is the least positive integer for which $x_n > 50$, then $\sum_{i=1}^{n} \left(\frac{1}{x_i}\right)$ is equal to :

(A) $\frac{1}{8}$ (B) 3 (C) $\frac{13}{8}$ (D) $\frac{13}{4}$

Open solution

**Q141.** JEE Main 2018 (Offline)

Let $a_1, a_2, a_3, \ldots\ldots\ldots, a_{49}$ be in A.P. such that

$\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$.

$a_1^2 + a_2^2 + \ldots\ldots\ldots + a_{17}^2 = 140m$, then m is equal to

(A) 33 (B) 66 (C) 68 (D) 34

Open solution

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**Q142.** JEE Main 2018 (Online) 15th April Evening Slot

If a, b, c are in A.P. and $a^2$, $b^2$, $c^2$ are in G.P. such that $a < b < c$ and $a + b + c = \frac{3}{4}$, then the value of a is :

(A) $\frac{1}{4} - \frac{1}{4\sqrt{2}}$ (B) $\frac{1}{4} - \frac{1}{3\sqrt{2}}$

(C) $\frac{1}{4} - \frac{1}{2\sqrt{2}}$ (D) $\frac{1}{4} - \frac{1}{\sqrt{2}}$

Open solution

**Q143.** JEE Main 2018 (Online) 15th April Morning Slot

If $x_1, x_2, \ldots, x_n$ and $\frac{1}{h_1}, \frac{1}{h_2}, \ldots, \frac{1}{h_n}$ are two A.P.s such that $x_3 = h_2 = 8$ and $x_8 = h_7 = 20$, then $x_5.h_{10}$ equals :

(A) 2560 (B) 2650 (C) 3200 (D) 1600

Open solution

**Q144.** JEE Main 2018 (Online) 15th April Morning Slot

If b is the first term of an infinite G.P. whose sum is five, then b lies in the interval :

(A) $(-\infty, -10]$ (B) $(-10, 0)$ (C) $(0, 10)$ (D) $[10, \infty)$

Open solution

**Q145.** JEE Main 2017 (Online) 9th April Morning Slot

If three positive numbers a, b and c are in A.P. such that abc = 8, then the minimum possible value of b is :

(A) 2 (B) $4^{\frac{1}{3}}$ (C) $4^{\frac{2}{3}}$ (D) 4

Open solution

**Q146.** JEE Main 2017 (Online) 8th April Morning Slot

If the arithmetic mean of two numbers a and b, $a > b > 0$, is five times their geometric mean, then $\frac{a+b}{a-b}$ is equal to :

(A) $\frac{\sqrt{6}}{2}$ (B) $\frac{3\sqrt{2}}{4}$ (C) $\frac{7\sqrt{3}}{12}$ (D) $\frac{5\sqrt{6}}{12}$

Open solution

**Q147.** JEE Main 2016 (Online) 10th April Morning Slot

Let $a_1, a_2, a_3, \ldots\ldots, a_n, \ldots\ldots$ be in A.P.

If $a_3 + a_7 + a_{11} + a_{15} = 72$,

then the sum of its first 17 terms is equal to :

(A) 306 (B) 153 (C) 612 (D) 204

Open solution

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**Q148.** JEE Main 2016 (Online) 10th April Morning Slot

If A > 0, B > 0 and A + B = $\frac{\pi}{6}$,

then the minimum value of tanA + tanB is :

(A) $\sqrt{3}-\sqrt{2}$ (B) $2-\sqrt{3}$ (C) $4-2\sqrt{3}$ (D) $\frac{2}{\sqrt{3}}$

Open solution

**Q149.** JEE Main 2016 (Online) 9th April Morning Slot

Let x, y, z be positive real numbers such that x + y + z = 12 and $x^3y^4z^5 = (0.1)\,(600)^3$. Then $x^3 + y^3 + z^3$ is equal to :

(A) 270 (B) 258 (C) 342 (D) 216

Open solution

**Q150.** JEE Main 2016 (Offline)

If the $2^{nd}, 5^{th}\ and\ 9^{th}$ terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is :

(A) 1 (B) $\frac{7}{4}$ (C) $\frac{8}{5}$ (D) $\frac{4}{3}$

Open solution

**Q151.** JEE Main 2015 (Online) 11th April Morning Slot

The sum of the 3rd and the 4th terms of a G.P. is 60 and the product of its first three terms is 1000. If the first term of this G.P. is positive, then its 7th term is :

(A) 7290 (B) 320 (C) 640 (D) 2430

Open solution

**Q152.** JEE Main 2015 (Online) 10th April Morning Slot

Let the sum of the first three terms of an A.P. be 39 and the sum of its last four terms be 178. If the first term of this A.P. is 10, then the median of the A.P. is :

(A) 26.5 (B) 28 (C) 29.5 (D) 31

Open solution

**Q153.** JEE Main 2015 (Offline)

If m is the A.M. of two distinct real numbers l and n $(l, n > 1)$ and $G_1, G_2$ and $G_3$ are three geometric means between $l$ and n, then $G_1^4 + 2G_2^4 + G_3^4$ equals:

(A) $4\,lmn^2$ (B) $4\,l^2m^2n^2$ (C) $4\,l^2m\,n$ (D) $4\,l\,m^2n$

Open solution

**Q154.** JEE Main 2014 (Online) 19th April Morning Slot

The number of terms in an A.P. is even; the sum of the odd terms in it is 24 and that of the even terms is 30. If the last term exceeds the first term by $10\frac{1}{2}$, then the number of terms in the A.P. is :

(A) 4 (B) 8 (C) 12 (D) 16

Open solution

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**Q155.** JEE Main 2014 (Online) 12th April Morning Slot

Let G be the geometric mean of two positive numbers a and b, and M be the arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}$.
If $\frac{1}{M}$ : G is 4 : 5, then a : b can be :

(A) 1 : 4 (B) 1 : 2 (C) 2 : 3 (D) 3 : 4

Open solution

**Q156.** JEE Main 2014 (Online) 12th April Morning Slot

The least positive integer n such that
$1-\frac{2}{3}-\frac{2}{3^2}-\ldots\ldots-\frac{2}{3^{n-1}}<\frac{1}{100}$, is :

(A) 4 (B) 5 (C) 6 (D) 7

Open solution

**Q157.** JEE Main 2014 (Online) 11th April Morning Slot

In a geometric progression, If the ratio of the sum of first 5 terms to the sum of their reciprocals is 49, and the sum of the first and the third term is 35. Then the first term of this geometric progression is :

(A) 7 (B) 21 (C) 28 (D) 42

Open solution

**Q158.** JEE Main 2014 (Online) 11th April Morning Slot

The sum of the first 20 terms common between the series 3 + 7 + 11 + 15 + . . . . . and 1 + 6 + 11 + 16 + . . . . . , is :

(A) 4000 (B) 4020 (C) 4200 (D) 4220

Open solution

**Q159.** JEE Main 2014 (Online) 9th April Morning Slot

Given an A.P. whose terms are all positive integers. The sum of the first nine terms is greater than 200 and less than 220. If the second term in it is 12, then its $4^{th}$ term is :

(A) 8 (B) 16 (C) 20 (D) 24

Open solution

**Q160.** JEE Main 2014 (Offline)

Three positive numbers form an increasing G.P. If the middle term in this G.P. is doubled, the new numbers are in A.P. then the common ratio of the G.P. is :

(A) $2-\sqrt{3}$ (B) $2+\sqrt{3}$ (C) $\sqrt{2}+\sqrt{3}$ (D) $3+\sqrt{2}$

Open solution

**Q161.** JEE Main 2013 (Online) 25th April Morning Slot

Given a sequence of 4 numbers, first three of which are in G.P and the last three are in A.P. with common difference six. If first and last terms of this sequence are equal, then the last term is

(A) 16 (B) 8 (C) 4 (D) 2

Open solution

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**Q162.** JEE Main 2013 (Online) 23th April Morning Slot

If $a_1, a_2, a_3, \ldots\ldots, a_n, \ldots.$ are in A.P, such that $a_4 - a_7 + a_{10} = m$, then the sum of first 13 terms of this A.P. is :

(A) 10m (B) 12m (C) 13m (D) 15m

Open solution

**Q163.** JEE Main 2013 (Online) 22th April Morning Slot

Given sum of the first n terms of an A.P. is $2n + 3n^2$. Another A.P. is formed with the same first term and double of the common difference, the sum of n terms of the new A.P. is:

(A) $n + 4n^2$ (B) $6n^2 - n$ (C) $n^2 + 4n$ (D) $3n + 2n^2$

Open solution

**Q164.** JEE Main 2013 (Online) 9th April Morning Slot

Let $a_1, a_2, a_3, \ldots.$ be an A.P, such that $\frac{a_1+a_2+\ldots.+a_p}{a_1+a_2+a_3+\ldots+a_q} = \frac{p^3}{q^3}$; $p \neq q$, then $\frac{a_6}{a_{21}}$ is equal to:

(A) $\frac{41}{11}$ (B) $\frac{31}{121}$ (C) $\frac{11}{41}$ (D) $\frac{121}{1681}$

Open solution

**Q165.** AIEEE 2012 (Online) 26th May

If the A.M. between $p^{th}$ and $q^{th}$ terms of an A.P. is equal to the A.M. between $r^{th}$ and $s^{th}$ terms of the same A.P., then $p + q$ is equal to

(A) $r + s - 1$ (B) $r + s - 2$ (C) $r + s + 1$ (D) $r + s$

Open solution

**Q166.** AIEEE 2012 (Online) 12th May

If $a, b, c, d$ and $p$ are distinct real numbers such that $(a^2 + b^2 + c^2)p^2 - 2p(ab + bc + cd) + (b^2 + c^2 + d^2) \leq 0$, then

(A) $a, b, c, d$ are in A.P. (B) $ab = cd$

(C) $ac = bd$ (D) $a, b, c, d$ are in G.P.

Open solution

**Q167.** AIEEE 2012 (Online) 7th May

The difference between the fourth term and the first term of a Geometrical Progression is 52. If the sum of its first three terms is 26 , then the sum of the first six terms of the progression is

(A) $63$ (B) $189$ (C) $728$ (D) $364$

Open solution

**Q168.** AIEEE 2012

If 100 times the 100th term of an AP with non zero common difference equals the 50 times its 50th term, then the 150th term of this AP is :

(A) $-150$ (B) 150 times its 50th term

(C) $150$ (D) zero

Open solution

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**Q169.** AIEEE 2011

A man saves ₹ 200 in each of the first three months of his service. In each of the subsequent months his saving increases by ₹ 40 more than the saving of immediately previous month. His total saving from the start of service will be ₹ 11040 after

(A) 19 months (B) 20 months (C) 21 months (D) 18 months

Open solution

**Q170.** AIEEE 2010

A person is to count 4500 currency notes. Let $a_n$ denote the number of notes he counts in the $n^{th}$ minute. If $a_1 = a_2 = ....= a_{10} = 150$ and $a_{10}, a_{11},....$ are in an AP with common difference - 2, then the time taken by him to count all notes is

(A) 34 minutes (B) 125 minutes (C) 135 minutes (D) 24 minutes

Open solution

**Q171.** AIEEE 2008

The first two terms of a geometric progression add up to 12. The sum of the third and the fourth terms is 48. If the terms of the geometric progression are alternately positive and negative, then the first term is

(A) - 4 (B) - 12 (C) 12 (D) 4

Open solution

**Q172.** AIEEE 2007

In a geometric progression consisting of positive terms, each term equals the sum of the next two terms. Then the common ratio of its progression equals

(A) $\sqrt{5}$ (B) $\frac{1}{2}\left(\sqrt{5}-1\right)$

(C) $\frac{1}{2}\left(1-\sqrt{5}\right)$ (D) $\frac{1}{2}\sqrt{5}$.

Open solution

**Q173.** AIEEE 2006

Let $a_1, a_2, a_3$.....be terms of an A.P. If $\frac{a_1+a_2+.....a_p}{a_1+a_2+.....a_q} = \frac{p^2}{q^2}, p \neq q, then \frac{a_6}{a_{21}}$ equals

(A) $\frac{41}{11}$ (B) $\frac{7}{2}$ (C) $\frac{2}{7}$ (D) $\frac{11}{41}$

Open solution

**Q174.** AIEEE 2006

If $a_1, a_2, \ldots . a_n$ are in H.P., then the expression $a_1\, a_2 + a_2\, a_3 + \ldots . + a_{n-1}\, a_n$ is equal to

(A) $n(a_1 - a_n)$ (B) $(n-1)(a_1 - a_n)$ (C) $na_1a_n$ (D) $(n-1)\, a_1a_n$

Open solution

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**Q175.** AIEEE 2005

If $x = \sum_{n=0}^{\infty} a^n,\ y = \sum_{n=0}^{\infty} b^n,\ z = \sum_{n=0}^{\infty} c^n,$ where a, b, c are in A.P and $|a| < 1,\ |b| < 1,\ |c| < 1$ then x, y, z are in

(A) G.P. (B) A.P.

(C) Arithmetic-Geometric Progression (D) H.P.

Open solution

**Q176.** AIEEE 2004

Let $T_r$ be the rth term of an A.P. whose first term is a and common difference is d. If for some positive integers m, n, $m \neq n,\ T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then a - d equals

(A) $\frac{1}{m} + \frac{1}{n}$ (B) 1 (C) $\frac{1}{mn}$ (D) 0

Open solution

**Q177.** AIEEE 2002

The sum of integers from 1 to 100 that are divisible by 2 or 5 is :

(A) 3000 (B) 3050 (C) 3600 (D) 3250

Open solution

**Q178.** AIEEE 2002

If 1, $\log_9 (3^{1-x} + 2),\ \log_3 (4 \cdot 3^x - 1)$ are in A.P. then x equals

(A) $\log_3 4$ (B) $1 - \log_3 4$ (C) $1 - \log_4 3$ (D) $\log_4 3$

Open solution

**Q179.** AIEEE 2002

l, m, n are the $p^{th}$, $q^{th}$ and $r^{th}$ term of a G.P all positive, $then \begin{vmatrix} \log l & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix} equals$

(A) - 1 (B) 2 (C) 1 (D) 0

Open solution

**Q180.** AIEEE 2002

Fifth term of a GP is 2, then the product of its first 9 terms is

(A) 256 (B) 512 (C) 1024 (D) none of these

Open solution

**Q181.** AIEEE 2002

Sum of infinite number of terms of GP is 20 and sum of their square is 100. The common ratio of GP is

(A) 5 (B) 3/5 (C) 8/5 (D) 1/5

Open solution

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## Numerical 49 questions

**Q182.** JEE Main 2026 (Online) 6th April Morning Shift

For the functions $f(\theta)=\alpha\tan^2\theta+\beta\cot^2\theta$, and $g(\theta)=\alpha\sin^2\theta+\beta\cos^2\theta, \alpha>\beta>0$, let $\min\limits_{0<\theta<\frac{\pi}{2}} f(\theta)=\max\limits_{0<\theta<\pi} g(\theta)$. If the first term of a G.P. is $\left(\frac{\alpha}{2\beta}\right)$, its common ratio is $\left(\frac{2\beta}{\alpha}\right)$ and the sum of its first 10 terms is $\frac{m}{n}, \gcd(m,n)=1$, then $m+n$ is equal to ____ .

Open solution

**Q183.** JEE Main 2026 (Online) 2nd April Morning Shift

If $\sum\limits_{k=1}^{n} a_k=6n^3$, then $\sum\limits_{k=1}^{6}\left(\frac{a_{k+1}-a_k}{36}\right)^2$ is equal to _____.

Open solution

**Q184.** JEE Main 2026 (Online) 28th January Evening Shift

If $\sum\limits_{r=1}^{25}\left(\frac{r}{r^4+r^2+1}\right)=\frac{p}{q}$, where p and q are positive integers such that $\gcd(p,q)=1$, then p + q is equal to _____.

Open solution

**Q185.** JEE Main 2026 (Online) 28th January Morning Shift

In a G.P., if the product of the first three terms is 27 and the set of all possible values for the sum of its first three terms is $\mathbb{R}-(a,b)$, then $a^2+b^2$ is equal to

____ .

Open solution

**Q186.** JEE Main 2026 (Online) 22nd January Evening Shift

Suppose $a, b, c$ are in A.P. and $a^2, 2b^2, c^2$ are in G.P. If $a<b<c$ and $a+b+c=1$, then $9\left(a^2+b^2+c^2\right)$ is equal to ____ .

Open solution

**Q187.** JEE Main 2026 (Online) 21st January Morning Shift

Let $a_1=1$ and for $n\geqslant 1, a_{n+1}=\frac{1}{2}a_n+\frac{n^2-2n-1}{n^2(n+1)^2}$. Then $\left|\sum_{n=1}^{\infty}\left(a_n-\frac{2}{n^2}\right)\right|$ is equal to ____ .

Open solution

**Q188.** JEE Main 2025 (Online) 2nd April Evening Shift

If the sum of the first 10 terms of the series $\frac{4\cdot1}{1+4\cdot1^4}+\frac{4\cdot2}{1+4\cdot2^4}+\frac{4\cdot3}{1+4\cdot3^4}+\ldots..$ is $\frac{m}{n}$, where $\gcd(m,n)=1$, then $m+n$ is equal to _____

Open solution

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**Q189.** JEE Main 2025 (Online) 29th January Evening Shift

Let $a_1, a_2, \ldots, a_{2024}$ be an Arithmetic Progression such that $a_1 + (a_5 + a_{10} + a_{15} + \ldots + a_{2020}) + a_{2024} = 2233$. Then $a_1 + a_2 + a_3 + \ldots + a_{2024}$ is equal to _____.

Open solution

**Q190.** JEE Main 2025 (Online) 28th January Evening Shift

The interior angles of a polygon with $n$ sides, are in an A.P. with common difference 6°. If the largest interior angle of the polygon is 219°, then $n$ is equal to _____.

Open solution

**Q191.** JEE Main 2025 (Online) 23rd January Evening Shift

The roots of the quadratic equation $3x^2 - px + q = 0$ are $10^{th}$ and $11^{th}$ terms of an arithmetic progression with common difference $\frac{3}{2}$. If the sum of the first 11 terms of this arithmetic progression is 88 , then $q - 2p$ is equal to _____ .

Open solution

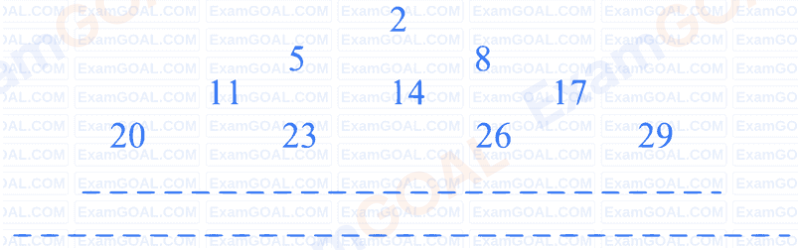
**Q192.** JEE Main 2024 (Online) 9th April Evening Shift

If $\left(\frac{1}{\alpha+1} + \frac{1}{\alpha+2} + \ldots\ldots + \frac{1}{\alpha+1012}\right) - \left(\frac{1}{2\cdot1} + \frac{1}{4\cdot3} + \frac{1}{6\cdot5} + \ldots\ldots + \frac{1}{2024\cdot2023}\right) = \frac{1}{2024}$, then $\alpha$ is equal to _____.

Open solution

**Q193.** JEE Main 2024 (Online) 8th April Evening Shift

An arithmetic progression is written in the following way




The sum of all the terms of the $10^{th}$ row is _____.

Open solution

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**Q194.** JEE Main 2024 (Online) 8th April Morning Shift

Let the positive integers be written in the form :




If the $k^{\text{th}}$ row contains exactly $k$ numbers for every natural number $k$, then the row in which the number 5310 will be, is _____.

Open solution

**Q195.** JEE Main 2024 (Online) 6th April Evening Shift

If $S(x) = (1+x) + 2(1+x)^2 + 3(1+x)^3 + \cdots + 60(1+x)^{60}, x \neq 0$, and $(60)^2\, S(60) = a(b)^b + b$, where $a, b \in N$, then $(a+b)$ equal to _____.

Open solution

**Q196.** JEE Main 2024 (Online) 6th April Morning Shift

Let the first term of a series be $T_1 = 6$ and its $r^{\text{th}}$ term $T_r = 3T_{r-1} + 6^r, r = 2, 3, \ldots\ldots\ldots n$. If the sum of the first $n$ terms of this series is $\frac{1}{5}\left(n^2 - 12n + 39\right)\left(4 \cdot 6^n - 5 \cdot 3^n + 1\right)$, then $n$ is equal to _____.

Open solution

**Q197.** JEE Main 2024 (Online) 5th April Evening Shift

If $1 + \frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}} + \frac{5-2\sqrt{6}}{18} + \frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}} + \frac{49-20\sqrt{6}}{180} + \ldots$ upto $\infty = 2 + \left(\sqrt{\frac{b}{a}} + 1\right)\log_e\left(\frac{a}{b}\right)$, where a and b are integers with $\gcd(a, b) = 1$, then $11a + 18b$ is equal to _____.

Open solution

**Q198.** JEE Main 2024 (Online) 5th April Morning Shift

Let $a_1, a_2, a_3, \ldots$ be in an arithmetic progression of positive terms.

Let $A_k = a_1^2 - a_2^2 + a_3^2 - a_4^2 + \ldots + a_{2k-1}^2 - a_{2k}^2$.

If $A_3 = -153$, $A_5 = -435$ and $a_1^2 + a_2^2 + a_3^2 = 66$, then $a_{17} - A_7$ is equal to _____.

Open solution

**Q199.** JEE Main 2024 (Online) 1st February Evening Shift

If three successive terms of a G.P. with common ratio $r(r > 1)$ are the lengths of the sides of a triangle and $[r]$ denotes the greatest integer less than or equal to $r$, then $3[r] + [-r]$ is equal to _____.

Open solution

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**Q200.** JEE Main 2024 (Online) 1st February Morning Shift

Let $3, 7, 11, 15, \ldots, 403$ and $2, 5, 8, 11, \ldots, 404$ be two arithmetic progressions. Then the sum, of the common terms in them, is equal to _____.

Open solution

**Q201.** JEE Main 2024 (Online) 30th January Evening Shift

Let $S_n$ be the sum to $n$-terms of an arithmetic progression $3, 7, 11, \ldots$. If $40 < \left(\frac{6}{n(n+1)} \sum_{k=1}^{n} S_k\right) < 42$, then $n$ equals _____.

Open solution

**Q202.** JEE Main 2024 (Online) 30th January Morning Shift

Let $\alpha = 1^2 + 4^2 + 8^2 + 13^2 + 19^2 + 26^2 + \ldots$ upto 10 terms and $\beta = \sum_{n=1}^{10} n^4$. If $4\alpha - \beta = 55k + 40$, then k is equal to _____.

Open solution

**Q203.** JEE Main 2024 (Online) 27th January Morning Shift

If $8 = 3 + \frac{1}{4}(3 + p) + \frac{1}{4^2}(3 + 2p) + \frac{1}{4^3}(3 + 3p) + \cdots\cdots\infty$, then the value of $p$ is _____.

Open solution

**Q204.** JEE Main 2023 (Online) 6th April Evening Shift

If

$$(20)^{19} + 2(21)(20)^{18} + 3(21)^2(20)^{17} + \ldots + 20(21)^{19} = k(20)^{19},$$

then $k$ is equal to _____.

Open solution

**Q205.** JEE Main 2023 (Online) 1st February Evening Shift

The sum of the common terms of the following three arithmetic progressions.

$3, 7, 11, 15, \ldots., 399$,

$2, 5, 8, 11, \ldots., 359$ and

$2, 7, 12, 17, \ldots., 197$,

is equal to _____.

Open solution

**Q206.** JEE Main 2023 (Online) 1st February Morning Shift

Let $a_1 = 8, a_2, a_3, \ldots, a_n$ be an A.P. If the sum of its first four terms is 50 and the sum of its last four terms is 170 , then the product of its middle two terms is _____.

Open solution

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**Q207.** JEE Main 2023 (Online) 31st January Morning Shift

Let $a_1, a_2, \ldots, a_n$ be in A.P. If $a_5 = 2a_7$ and $a_{11} = 18$, then

$12\left(\frac{1}{\sqrt{a_{10}}+\sqrt{a_{11}}} + \frac{1}{\sqrt{a_{11}}+\sqrt{a_{12}}} + \ldots + \frac{1}{\sqrt{a_{17}}+\sqrt{a_{18}}}\right)$ is equal to _____.

Open solution

**Q208.** JEE Main 2023 (Online) 29th January Evening Shift

Let $\{a_k\}$ and $\{b_k\}, k \in N$, be two G.P.s with common ratios $r_1$ and $r_2$ respectively such that $a_1 = b_1 = 4$ and $r_1 < r_2$. Let $c_k = a_k + b_k, k \in N$. If $c_2 = 5$ and $c_3 = \frac{13}{4}$ then $\sum_{k=1}^{\infty} c_k - (12a_6 + 8b_4)$ is equal to _____.

Open solution

**Q209.** JEE Main 2023 (Online) 29th January Morning Shift

Let $a_1, a_2, a_3, \ldots$ be a $GP$ of increasing positive numbers. If the product of fourth and sixth terms is 9 and the sum of fifth and seventh terms is 24, then $a_1a_9 + a_2a_4a_9 + a_5 + a_7$ is equal to _____.

Open solution

**Q210.** JEE Main 2023 (Online) 25th January Evening Shift

For the two positive numbers $a, b$, if $a, b$ and $\frac{1}{18}$ are in a geometric progression, while $\frac{1}{a}, 10$ and $\frac{1}{b}$ are in an arithmetic progression, then $16a + 12b$ is equal to _____.

Open solution

**Q211.** JEE Main 2023 (Online) 24th January Morning Shift

The $4^{\text{th}}$ term of GP is 500 and its common ratio is $\frac{1}{m}, m \in \mathbb{N}$. Let $S_n$ denote the sum of the first n terms of this GP. If $S_6 > S_5 + 1$ and $S_7 < S_6 + \frac{1}{2}$, then the number of possible values of m is _____

Open solution

**Q212.** JEE Main 2022 (Online) 29th July Morning Shift

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $\sum_{r=1}^{\infty} \frac{a_r}{2^r} = 4$, then $4a_2$ is equal to _____.

Open solution

**Q213.** JEE Main 2022 (Online) 26th July Evening Shift

Different A.P.'s are constructed with the first term 100, the last term 199, and integral common differences. The sum of the common differences of all such A.P.'s having at least 3 terms and at most 33 terms is _____.

Open solution

**Q214.** JEE Main 2022 (Online) 25th July Morning Shift

Let $a, b$ be two non-zero real numbers. If $p$ and $r$ are the roots of the equation $x^2 - 8ax + 2a = 0$ and q and s are the roots of the equation $x^2 + 12\,bx + 6\,b = 0$, such that $\frac{1}{p}, \frac{1}{q}, \frac{1}{r}, \frac{1}{s}$ are in A.P., then $a^{-1} - b^{-1}$ is equal to _____.

Open solution

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**Q215.** JEE Main 2022 (Online) 29th June Evening Shift

Let 3, 6, 9, 12, ..... upto 78 terms and 5, 9, 13, 17, ..... upto 59 terms be two series. Then, the sum of the terms common to both the series is equal to _____.

Open solution

**Q216.** JEE Main 2022 (Online) 28th June Evening Shift

Let for n = 1, 2, ....., 50, $S_n$ be the sum of the infinite geometric progression whose first term is $n^2$ and whose common ratio is $\frac{1}{(n+1)^2}$. Then the value of

$\frac{1}{26} + \sum_{n=1}^{50} \left(S_n + \frac{2}{n+1} - n - 1\right)$ is equal to _____.

Open solution

**Q217.** JEE Main 2022 (Online) 28th June Morning Shift

Let A = {1, $a_1$, $a_2$ ..... $a_{18}$, 77} be a set of integers with 1 < $a_1$ < $a_2$ < ..... < $a_{18}$ < 77.

Let the set A + A = {x + y : x, y $\in$ A} contain exactly 39 elements. Then, the value of $a_1 + a_2 +$ ..... + $a_{18}$ is equal to _____.

Open solution

**Q218.** JEE Main 2022 (Online) 26th June Evening Shift

If $a_1$ (> 0), $a_2$, $a_3$, $a_4$, $a_5$ are in a G.P., $a_2 + a_4 = 2a_3 + 1$ and $3a_2 + a_3 = 2a_4$, then $a_2 + a_4 + 2a_5$ is equal to _____.

Open solution

**Q219.** JEE Main 2021 (Online) 26th August Evening Shift

The sum of all 3-digit numbers less than or equal to 500, that are formed without using the digit "1" and they all are multiple of 11, is _____.

Open solution

**Q220.** JEE Main 2021 (Online) 26th August Evening Shift

Let $a_1$, $a_2$, ....., $a_{10}$ be an AP with common difference $-3$ and $b_1$, $b_2$, ....., $b_{10}$ be a GP with common ratio 2. Let $c_k = a_k + b_k$, k = 1, 2, ....., 10. If $c_2 = 12$ and $c_3 = 13$, then $\sum_{k=1}^{10} c_k$ is equal to _____.

Open solution

**Q221.** JEE Main 2021 (Online) 27th July Morning Shift

If $\log_3 2, \log_3(2^x - 5), \log_3\left(2^x - \frac{7}{2}\right)$ are in an arithmetic progression, then the value of x is equal to _____.

Open solution

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**Q222.** JEE Main 2021 (Online) 22th July Evening Shift

The sum of all the elements in the set {n$\in$ {1, 2, ....., 100} | H.C.F. of n and 2040 is 1} is equal to _____.

Open solution

**Q223.** JEE Main 2021 (Online) 16th March Evening Shift

Let $\frac{1}{16}$, a and b be in G.P. and $\frac{1}{a}$, $\frac{1}{b}$, 6 be in A.P., where a, b > 0. Then 72(a + b) is equal to _____.

Open solution

**Q224.** JEE Main 2021 (Online) 16th March Morning Shift

Consider an arithmetic series and a geometric series having four initial terms from the set {11, 8, 21, 16, 26, 32, 4}. If the last terms of these series are the maximum possible four digit numbers, then the number of common terms in these two series is equal to _____.

Open solution

**Q225.** JEE Main 2021 (Online) 26th February Evening Shift

If the arithmetic mean and geometric mean of the $p^{th}$ and $q^{th}$ terms of the sequence $-16, 8, -4, 2, .....$ satisfy the equation $4x^2 - 9x + 5 = 0$, then p + q is equal to _____.

Open solution

**Q226.** JEE Main 2021 (Online) 25th February Morning Shift

Let $A_1, A_2, A_3, .....$ be squares such that for each $n \geq 1$, the length of the side of $A_n$ equals the length of diagonal of $A_{n+1}$. If the length of the side of $A_1$ is 12 cm, then the smallest value of n for which area of $A_n$ is less than one, is _____.

Open solution

**Q227.** JEE Main 2021 (Online) 24th February Evening Shift

The sum of first four terms of a geometric progression (G. P.) is $\frac{65}{12}$ and the sum of their respective reciprocals is $\frac{65}{18}$. If the product of first three terms of the G.P. is 1, and the third term is $\alpha$, then $2\alpha$ is _____.

Open solution

**Q228.** JEE Main 2020 (Online) 3rd September Evening Slot

If m arithmetic means (A.Ms) and three geometric means (G.Ms) are inserted between 3 and 243 such that $4^{th}$ A.M. is equal to $2^{nd}$ G.M., then m is equal to _____ .

Open solution

**Q229.** JEE Main 2020 (Online) 3rd September Morning Slot

The value of $(0.16)^{\log_{2.5}\left(\frac{1}{3}+\frac{1}{3^2}+....to\ \infty\right)}$ is equal to _____.

Open solution

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**Q230.** JEE Main 2020 (Online) 9th January Evening Slot

The number of terms common to the two A.P.'s 3, 7, 11, ....., 407 and 2, 9, 16, ....., 709 is _____.

Open solution

## Out of syllabus 111 questions

*These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.*

**Q231.** JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

Let $< a_n >$ be a sequence such that $a_1 + a_2 + \ldots + a_n = \frac{n^2+3n}{(n+1)(n+2)}$. If $28 \sum_{k=1}^{10} \frac{1}{a_k} = p_1 p_2 p_3 \ldots p_m$, where $p_1, p_2, \ldots, p_m$ are the first $m$ prime numbers, then $m$ is equal to

(A) 5 (B) 7 (C) 6 (D) 8

Open solution

**Q232.** JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

If $S_n = 4 + 11 + 21 + 34 + 50 + \ldots$ to $n$ terms, then $\frac{1}{60}(S_{29} - S_9)$ is equal to :

(A) 227 (B) 226 (C) 220 (D) 223

Open solution

**Q233.** JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Let $a_n$ be the $n^{th}$ term of the series $5 + 8 + 14 + 23 + 35 + 50 + \ldots$ and $S_n = \sum_{k=1}^{n} a_k$. Then $S_{30} - a_{40}$ is equal to :

(A) 11280 (B) 11290 (C) 11310 (D) 11260

Open solution

**Q234.** JEE Main 2023 (Online) 8th April Morning Shift · MCQ (Single Correct Answer)

Let $S_K = \frac{1+2+\ldots+K}{K}$ and $\sum_{j=1}^{n} S_j^2 = \frac{n}{A}(Bn^2 + Cn + D)$, where $A, B, C, D \in \mathbb{N}$ and $A$ has least value.

Then

(A) $A + B + C + D$ is divisible by 5 (B) $A + C + D$ is not divisible by $B$

(C) $A + B = 5(D - C)$ (D) $A + B$ is divisible by D

Open solution

**Q235.** JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

If $\gcd(m, n) = 1$ and $1^2 - 2^2 + 3^2 - 4^2 + \ldots . + (2021)^2 - (2022)^2 + (2023)^2 = 1012\, m^2 n$ then $m^2 - n^2$ is equal to :

(A) 220 (B) 200 (C) 240 (D) 180

Open solution

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**Q236.** JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

The sum of the first 20 terms of the series $5 + 11 + 19 + 29 + 41 + \ldots$ is :

(A) 3420 (B) 3450 (C) 3250 (D) 3520

Open solution

**Q237.** JEE Main 2023 (Online) 1st February Evening Shift · MCQ (Single Correct Answer)

The sum $\sum_{n=1}^{\infty} \frac{2n^2+3n+4}{(2n)!}$ is equal to :

(A) $\frac{11e}{2} + \frac{7}{2e}$ (B) $\frac{13e}{4} + \frac{5}{4e} - 4$

(C) $\frac{11e}{2} + \frac{7}{2e} - 4$ (D) $\frac{13e}{4} + \frac{5}{4e}$

Open solution

**Q238.** JEE Main 2023 (Online) 1st February Morning Shift · MCQ (Single Correct Answer)

The sum of 10 terms of the series

$\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \ldots.$ is

(A) $\frac{58}{111}$ (B) $\frac{56}{111}$ (C) $\frac{55}{111}$ (D) $\frac{59}{111}$

Open solution

**Q239.** JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

If $a_n = \frac{-2}{4n^2-16n+15}$, then $a_1 + a_2 + \ldots. + a_{25}$ is equal to :

(A) $\frac{51}{144}$ (B) $\frac{49}{138}$ (C) $\frac{50}{141}$ (D) $\frac{52}{147}$

Open solution

**Q240.** JEE Main 2022 (Online) 30th July Evening Shift · MCQ (Single Correct Answer)

$2\sum_{n=2}^{\infty} \frac{n(2n^2+3)}{(n+1)!}$

is equal to

(A) $2(e+7)$ (B) $2e+1$ (C) $2e+9$ (D) $2(e+1)$

Open solution

**Q241.** JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence such that $a_0 = a_1 = 0$ and $a_{n+2} = 3a_{n+1} - 2a_n + 1, \forall n \geq 0.$

Then $a_{25}a_{23} - 2a_{25}a_{22} - 2a_{23}a_{24} + 4a_{22}a_{24}$ is equal to

(A) 483 (B) 528 (C) 575 (D) 624

Open solution

**Q242.** JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

$\sum_{r=1}^{20} (r^2+1)(r!)$ is equal to

(A) $22! - 21!$ (B) $22! - 2(21!)$ (C) $21! - 2(20!)$ (D) $21! - 20!$

Open solution

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**Q243.** JEE Main 2022 (Online) 29th July Morning Shift · MCQ (Single Correct Answer)

If $\frac{1}{(20-a)(40-a)}+\frac{1}{(40-a)(60-a)}+\ldots+\frac{1}{(180-a)(200-a)}=\frac{1}{256}$, then the maximum value of a is :

(A) 198 (B) 202 (C) 212 (D) 218

Open solution

**Q244.** JEE Main 2022 (Online) 28th July Morning Shift · MCQ (Single Correct Answer)

Consider the sequence $a_1, a_2, a_3, \ldots$ such that $a_1=1, a_2=2$ and $a_{n+2}=\frac{2}{a_{n+1}}+a_n$ for $n=1,2,3,\ldots$. If $\left(\frac{a_1+\frac{1}{a_2}}{a_3}\right)\cdot\left(\frac{a_2+\frac{1}{a_3}}{a_4}\right)\cdot\left(\frac{a_3+\frac{1}{a_4}}{a_5}\right)\ldots\left(\frac{a_{30}+\frac{1}{a_{31}}}{a_{32}}\right)=2^{\alpha}\left({}^{61}C_{31}\right)$, then $\alpha$ is equal to :

(A) $-30$ (B) $-31$ (C) $-60$ (D) $-61$

Open solution

**Q245.** JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

The sum $\sum_{n=1}^{21}\frac{3}{(4n-1)(4n+3)}$ is equal to

(A) $\frac{7}{87}$ (B) $\frac{7}{29}$ (C) $\frac{14}{87}$ (D) $\frac{21}{29}$

Open solution

**Q246.** JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

The value of $1+\frac{1}{1+2}+\frac{1}{1+2+3}+\ldots.+\frac{1}{1+2+3+\ldots..+11}$ is equal to:

(A) $\frac{20}{11}$ (B) $\frac{11}{6}$ (C) $\frac{241}{132}$ (D) $\frac{21}{11}$

Open solution

**Q247.** JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

The sum of the infinite series $1+\frac{5}{6}+\frac{12}{6^2}+\frac{22}{6^3}+\frac{35}{6^4}+\frac{51}{6^5}+\frac{70}{6^6}+\ldots..$ is equal to :

(A) $\frac{425}{216}$ (B) $\frac{429}{216}$ (C) $\frac{288}{125}$ (D) $\frac{280}{125}$

Open solution

**Q248.** JEE Main 2022 (Online) 29th June Morning Shift · MCQ (Single Correct Answer)

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence such that $a_0=a_1=0$ and $a_{n+2}=2a_{n+1}-a_n+1$ for all $n\geq 0$. Then, $\sum_{n=2}^{\infty}\frac{a_n}{7^n}$ is equal to:

(A) $\frac{6}{343}$ (B) $\frac{7}{216}$ (C) $\frac{8}{343}$ (D) $\frac{49}{216}$

Open solution

**Q249.** JEE Main 2022 (Online) 27th June Evening Shift · MCQ (Single Correct Answer)

Let $S=2+\frac{6}{7}+\frac{12}{7^2}+\frac{20}{7^3}+\frac{30}{7^4}+\ldots..$ Then 4S is equal to

(A) $\left(\frac{7}{3}\right)^2$ (B) $\frac{7^3}{3^2}$ (C) $\left(\frac{7}{3}\right)^3$ (D) $\frac{7^2}{3^3}$

Open solution

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**Q250.** JEE Main 2022 (Online) 26th June Evening Shift · MCQ (Single Correct Answer)

If $A = \sum_{n=1}^{\infty} \frac{1}{(3+(-1)^n)^n}$ and $B = \sum_{n=1}^{\infty} \frac{(-1)^n}{(3+(-1)^n)^n}$, then $\frac{A}{B}$ is equal to :

(A) $\frac{11}{9}$ (B) 1 (C) $-\frac{11}{9}$ (D) $-\frac{11}{3}$

Open solution

**Q251.** JEE Main 2022 (Online) 25th June Evening Shift · MCQ (Single Correct Answer)

The sum $1 + 2 \cdot 3 + 3 \cdot 3^2 + \ldots\ldots + 10 \cdot 3^9$ is equal to :

(A) $\frac{2 \cdot 3^{12}+10}{4}$ (B) $\frac{19 \cdot 3^{10}+1}{4}$ (C) $5 \cdot 3^{10} - 2$ (D) $\frac{9 \cdot 3^{10}+1}{2}$

Open solution

**Q252.** JEE Main 2021 (Online) 1st September Evening Shift · MCQ (Single Correct Answer)

Let $S_n = 1 \cdot (n-1) + 2 \cdot (n-2) + 3 \cdot (n-3) + \ldots\ldots + (n-1) \cdot 1$, $n \geq 4$.

The sum $\sum_{n=4}^{\infty} \left( \frac{2S_n}{n!} - \frac{1}{(n-2)!} \right)$ is equal to :

(A) $\frac{e-1}{3}$ (B) $\frac{e-2}{6}$ (C) $\frac{e}{3}$ (D) $\frac{e}{6}$

Open solution

**Q253.** JEE Main 2021 (Online) 31st August Morning Shift · MCQ (Single Correct Answer)

The sum of 10 terms of the series

$\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 \times 4^2} + \ldots\ldots$ is :

(A) 1 (B) $\frac{120}{121}$ (C) $\frac{99}{100}$ (D) $\frac{143}{144}$

Open solution

**Q254.** JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

If $0 < x < 1$ and $y = \frac{1}{2}x^2 + \frac{2}{3}x^3 + \frac{3}{4}x^4 + \ldots\ldots$, then the value of $e^{1+y}$ at $x = \frac{1}{2}$ is :

(A) $\frac{1}{2}e^2$ (B) $2e$ (C) $\frac{1}{2}\sqrt{e}$ (D) $2e^2$

Open solution

**Q255.** JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

If $0 < x < 1$, then $\frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 + \ldots\ldots$, is equal to :

(A) $x\left(\frac{1+x}{1-x}\right) + \log_e(1-x)$ (B) $x\left(\frac{1-x}{1+x}\right) + \log_e(1-x)$

(C) $\frac{1-x}{1+x} + \log_e(1-x)$ (D) $\frac{1+x}{1-x} + \log_e(1-x)$

Open solution

**Q256.** JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

If for $x, y \in R$, $x > 0$, $y = \log_{10}x + \log_{10}x^{1/3} + \log_{10}x^{1/9} + \ldots\ldots$ upto $\infty$ terms

and $\frac{2+4+6+\ldots.+2y}{3+6+9+\ldots.+3y} = \frac{4}{\log_{10}x}$, then the ordered pair (x, y) is equal to :

(A) $(10^6, 6)$ (B) $(10^4, 6)$ (C) $(10^2, 3)$ (D) $(10^6, 9)$

Open solution

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**Q257.** JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

The sum of the series

$\frac{1}{x+1}+\frac{2}{x^2+1}+\frac{2^2}{x^4+1}+\ldots\ldots+\frac{2^{100}}{x^{2^{100}}+1}$ when x = 2 is :

(A) $1+\frac{2^{101}}{4^{101}-1}$ (B) $1+\frac{2^{100}}{4^{101}-1}$

(C) $1-\frac{2^{100}}{4^{100}-1}$ (D) $1-\frac{2^{101}}{2^{400}-1}$

Open solution

**Q258.** JEE Main 2021 (Online) 20th July Evening Shift · MCQ (Single Correct Answer)

If sum of the first 21 terms of the series $\log_{9^{1/2}}x+\log_{9^{1/3}}x+\log_{9^{1/4}}x+\ldots\ldots$, where x > 0 is 504, then x is equal to

(A) 243 (B) 9 (C) 7 (D) 81

Open solution

**Q259.** JEE Main 2021 (Online) 18th March Morning Shift · MCQ (Single Correct Answer)

If $\alpha$, $\beta$ are natural numbers such that
$100^{\alpha}-199\beta$ = (100)(100) + (99)(101) + (98)(102) + ..... + (1)(199), then the slope of the line passing through ($\alpha$, $\beta$) and origin is :

(A) 540 (B) 550 (C) 530 (D) 510

Open solution

**Q260.** JEE Main 2021 (Online) 18th March Morning Shift · MCQ (Single Correct Answer)

$\frac{1}{3^2-1}+\frac{1}{5^2-1}+\frac{1}{7^2-1}+\ldots.+\frac{1}{(201)^2-1}$ is equal to

(A) $\frac{101}{404}$ (B) $\frac{25}{101}$ (C) $\frac{101}{408}$ (D) $\frac{99}{400}$

Open solution

**Q261.** JEE Main 2021 (Online) 26th February Evening Shift · MCQ (Single Correct Answer)

The sum of the series

$\sum_{n=1}^{\infty}\frac{n^2+6n+10}{(2n+1)!}$ is equal to :

(A) $\frac{41}{8}e+\frac{19}{8}e^{-1}-10$ (B) $\frac{41}{8}e-\frac{19}{8}e^{-1}-10$

(C) $\frac{41}{8}e+\frac{19}{8}e^{-1}+10$ (D) $-\frac{41}{8}e+\frac{19}{8}e^{-1}-10$

Open solution

**Q262.** JEE Main 2021 (Online) 26th February Morning Shift · MCQ (Single Correct Answer)

The sum of the infinite series
$1+\frac{2}{3}+\frac{7}{3^2}+\frac{12}{3^3}+\frac{17}{3^4}+\frac{22}{3^5}+\ldots\ldots$ is equal to :

(A) $\frac{9}{4}$ (B) $\frac{13}{4}$ (C) $\frac{15}{4}$ (D) $\frac{11}{4}$

Open solution

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**Q263.** JEE Main 2021 (Online) 25th February Morning Shift · MCQ (Single Correct Answer)

If $0 < \theta, \phi < \frac{\pi}{2}, x = \sum_{n=0}^{\infty} \cos^{2n}\theta, y = \sum_{n=0}^{\infty} \sin^{2n}\phi$ and $z = \sum_{n=0}^{\infty} \cos^{2n}\theta . \sin^{2n}\phi$ then :

(A) xy − z = (x + y)z

(B) xyz = 4

(C) xy + z = (x + y)z

(D) xy + yz + zx = z

Open solution

**Q264.** JEE Main 2020 (Online) 5th September Evening Slot · MCQ (Single Correct Answer)

If the sum of the first 20 terms of the series
$\log_{(7^{1/2})} x + \log_{(7^{1/3})} x + \log_{(7^{1/4})} x + \ldots$ is 460,
then x is equal to :

(A) $e^2$

(B) $7^{1/2}$

(C) $7^2$

(D) $7^{46/21}$

Open solution

**Q265.** JEE Main 2020 (Online) 5th September Morning Slot · MCQ (Single Correct Answer)

If $2^{10} + 2^9.3^1 + 2^8.3^2 + \ldots + 2.3^9 + 3^{10} = S - 2^{11}$, then S is equal to :

(A) $\frac{3^{11}}{2} + 2^{10}$

(B) $3^{11} - 2^{12}$

(C) $2.3^{11}$

(D) $3^{11}$

Open solution

**Q266.** JEE Main 2020 (Online) 4th September Morning Slot · MCQ (Single Correct Answer)

If $1+(1-2^2.1)+(1-4^2.3)+(1-6^2.5)+\ldots+(1-20^2.19) = \alpha - 220\beta$,
then an ordered pair $(\alpha, \beta)$ is equal to:

(A) (11, 103)

(B) (10, 103)

(C) (10, 97)

(D) (11, 97)

Open solution

**Q267.** JEE Main 2020 (Online) 3rd September Evening Slot · MCQ (Single Correct Answer)

If the sum of the series

$20 + 19\frac{3}{5} + 19\frac{1}{5} + 18\frac{4}{5} + \ldots$

upto nth term is 488 and the $n^{th}$ term is negative, then :

(A) n = 41

(B) n = 60

(C) $n^{th}$ term is −4

(D) $n^{th}$ term is $-4\frac{2}{5}$

Open solution

**Q268.** JEE Main 2020 (Online) 2nd September Evening Slot · MCQ (Single Correct Answer)

Let S be the sum of the first 9 terms of the series :
$\{x + ka\} + \{x^2 + (k + 2)a\} + \{x^3 + (k + 4)a\}$
$+ \{x^4 + (k + 6)a\} + \ldots$ where $a \neq 0$ and $x \neq 1$.

If $S = \frac{x^{10} - x + 45a(x-1)}{x-1}$, then k is equal to :

(A) -3

(B) 1

(C) -5

(D) 3

Open solution

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**Q269.** JEE Main 2020 (Online) 2nd September Morning Slot · MCQ (Single Correct Answer)

If |x| < 1, |y| < 1 and x ≠ y, then the sum to infinity of the following series

$(x + y) + (x^2+xy+y^2) + (x^3+x^2y + xy^2+y^3) + ....$

(A) $\frac{x+y-xy}{(1+x)(1+y)}$ (B) $\frac{x+y-xy}{(1-x)(1-y)}$

(C) $\frac{x+y+xy}{(1+x)(1+y)}$ (D) $\frac{x+y+xy}{(1-x)(1-y)}$

Open solution

**Q270.** JEE Main 2020 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

The product $2^{\frac{1}{4}}.4^{\frac{1}{16}}.8^{\frac{1}{48}}.16^{\frac{1}{128}}$ ... to $\infty$ is equal to :

(A) $2^{\frac{1}{4}}$ (B) $2^{\frac{1}{2}}$ (C) 1 (D) 2

Open solution

**Q271.** JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

If the sum of the first 40 terms of the series,
3 + 4 + 8 + 9 + 13 + 14 + 18 + 19 + ..... is (102)m, then m is equal to :

(A) 20 (B) 5 (C) 10 (D) 25

Open solution

**Q272.** JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

For x $\varepsilon$ R, let [x] denote the greatest integer $\leq$ x, then the sum of the series $\left[-\frac{1}{3}\right] + \left[-\frac{1}{3} - \frac{1}{100}\right] + \left[-\frac{1}{3} - \frac{2}{100}\right] + \ldots + \left[-\frac{1}{3} - \frac{99}{100}\right]$ is :

(A) - 153 (B) - 135 (C) - 133 (D) - 131

Open solution

**Q273.** JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

The sum
$1 + \frac{1^3+2^3}{1+2} + \frac{1^3+2^3+3^3}{1+2+3} + \ldots\ldots + \frac{1^3+2^3+3^3+\ldots+15^3}{1+2+3+\ldots+15} - \frac{1}{2}(1 + 2 + 3 + \ldots + 15)$ is equal to :

(A) 620 (B) 1240 (C) 1860 (D) 660

Open solution

**Q274.** JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The sum
$\frac{3\times1^3}{1^2} + \frac{5\times(1^3+2^3)}{1^2+2^2} + \frac{7\times(1^3+2^3+3^3)}{1^2+2^2+3^2} + \ldots\ldots$ upto 10 terms is:

(A) 600 (B) 660 (C) 680 (D) 620

Open solution

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**Q275.** JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

Some identical balls are arranged in rows to form an equilateral triangle. The first row consists of one ball, the second row consists of two balls and so on. If 99 more identical balls are addded to the total number of balls used in forming the equilaterial triangle, then all these balls can be arranged in a square whose each side contains exactly 2 balls less than the number of balls each side of the triangle contains. Then the number of balls used to form the equilateral triangle is :-

(A) 262 (B) 190 (C) 157 (D) 225

Open solution

**Q276.** JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

The sum of the series 1 + 2 × 3 + 3 × 5 + 4 × 7 +.... upto 11th term is :-

(A) 945 (B) 916 (C) 915 (D) 946

Open solution

**Q277.** JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

The sum $\sum_{k=1}^{20} k\frac{1}{2^k}$ is equal to

(A) $2-\frac{11}{2^{19}}$ (B) $2-\frac{3}{2^{17}}$ (C) $1-\frac{11}{2^{20}}$ (D) $2-\frac{21}{2^{20}}$

Open solution

**Q278.** JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

If the sum of the first 15 terms of the series $\left(\frac{3}{4}\right)^3+\left(1\frac{1}{2}\right)^3+\left(2\frac{1}{4}\right)^3+3^3+\left(3\frac{3}{4}\right)^3+\ldots.$ is equal to 225 k, then k is equal to :

(A) 9 (B) 108 (C) 27 (D) 54

Open solution

**Q279.** JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

Let $S_k = \frac{1+2+3+....+k}{k}$. If $S_1^2+S_2^2+.....+S_{10}^2=\frac{5}{12}A$, then A is equal to :

(A) 283 (B) 156 (C) 301 (D) 303

Open solution

**Q280.** JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

The sum of the following series

$$1+6+\frac{9\left(1^2+2^2+3^2\right)}{7}+\frac{12\left(1^2+2^2+3^2+4^2\right)}{9}$$

$$+\frac{15\left(1^2+2^2+...+5^2\right)}{11}+.....\text{ up to 15 terms, is :}$$

(A) 7520 (B) 7510 (C) 7830 (D) 7820

Open solution

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**Q281.** JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

The sum of the first 20 terms of the series

$1+\frac{3}{2}+\frac{7}{4}+\frac{15}{8}+\frac{31}{16}+\ldots$, is :

(A) $38+\frac{1}{2^{19}}$ (B) $38+\frac{1}{2^{20}}$ (C) $39+\frac{1}{2^{20}}$ (D) $39+\frac{1}{2^{19}}$

Open solution

**Q282.** JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

Let A be the sum of the first 20 terms and B be the sum of the first 40 terms of the series
$1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2$ .....
If B - 2A = 100$\lambda$, then $\lambda$ is equal to

(A) 496 (B) 232 (C) 248 (D) 464

Open solution

**Q283.** JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

Let $A_n = \left(\frac{3}{4}\right) - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 - \ldots\ldots + (-1)^{n-1}\left(\frac{3}{4}\right)^n$ and $B_n = 1 - A_n$.
Then, the least odd natural number p, so that $B_n > A_n$ , for all $n \geq p$, is :

(A) 9 (B) 7 (C) 11 (D) 5

Open solution

**Q284.** JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

Let

$S_n = \frac{1}{1^3}+\frac{1+2}{1^3+2^3}+\frac{1+2+3}{1^3+2^3+3^3}+\ldots\ldots\ldots+\frac{1+2+\ldots\ldots+n}{1^3+2^3+\ldots\ldots+n^3}$.

If 100 $S_n$ = n, then n is equal to :

(A) 199 (B) 99 (C) 200 (D) 19

Open solution

**Q285.** JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

If the sum of the first n terms of the series $\sqrt{3}+\sqrt{75}+\sqrt{243}+\sqrt{507}+\ldots\ldots$ is $435\sqrt{3}$, then n equals :

(A) 18 (B) 15 (C) 13 (D) 29

Open solution

**Q286.** JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

For any three positive real numbers a, b and c,

$9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$.
Then

(A) b, c and $a$ are in G.P. (B) b, c and $a$ are in A.P.
(C) $a$, b and c are in A.P. (D) $a$, b and c are in G.P.

Open solution

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**Q287.** JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

Let z = 1 + ai be a complex number, a > 0, such that $z^3$ is a real number.

Then the sum 1 + z + $z^2$ + . . . . .+ $z^{11}$ is equal to :

(A) $-1250\sqrt{3}\,i$ (B) $1250\sqrt{3}\,i$ (C) $1365\sqrt{3}i$ (D) $-1365\sqrt{3}i$

Open solution

**Q288.** JEE Main 2016 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The value of $\sum_{r=1}^{15} r^2 \left(\frac{^{15}C_r}{^{15}C_{r-1}}\right)$ is equal to :

(A) 560 (B) 680 (C) 1240 (D) 1085

Open solution

**Q289.** JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

If the sum of the first ten terms of the series $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \ldots\ldots\ldots is\ \frac{16}{5}m$, then m is equal to :

(A) 100 (B) 99 (C) 102 (D) 101

Open solution

**Q290.** JEE Main 2015 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

If $\sum_{n=1}^{5} \frac{1}{n(n+1)(n+2)(n+3)} = \frac{k}{3}$, then k is equal to :

(A) $\frac{55}{336}$ (B) $\frac{17}{105}$ (C) $\frac{1}{6}$ (D) $\frac{19}{112}$

Open solution

**Q291.** JEE Main 2015 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The value of $\sum_{r=16}^{30} (r+2)(r-3)$ is equal to :

(A) 7785 (B) 7780 (C) 7775 (D) 7770

Open solution

**Q292.** JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The sum of first 9 terms of the series.

$\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \ldots\ldots$

(A) 142 (B) 192 (C) 71 (D) 96

Open solution

**Q293.** JEE Main 2014 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If the sum

$\frac{3}{1^2} + \frac{5}{1^2+2^2} + \frac{7}{1^2+2^2+3^2} + \ldots\ldots +$

upto 20 terms is equal to $\frac{k}{21}$, then k is equal to :

(A) 120 (B) 180 (C) 240 (D) 60

Open solution

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**Q294.** JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

If $(10)^9 + 2(11)^1(10^8) + 3(11)^2(10)^7 + \ldots\ldots\ldots + 10(11)^9 = k(10)^9$, then k is equal to :

(A) 100 (B) 110 (C) $\frac{121}{10}$ (D) $\frac{441}{100}$

Open solution

**Q295.** JEE Main 2013 (Online) 25th April Morning Slot · MCQ (Single Correct Answer)

The value of $1^2 + 3^2 + 5^2 + \ldots\ldots\ldots + 25^2$ is:

(A) 2925 (B) 1469 (C) 1728 (D) 1456

Open solution

**Q296.** JEE Main 2013 (Online) 23th April Morning Slot · MCQ (Single Correct Answer)

The sum of the series :
$(2)^2 + 2(4)^2 + 3(6)^2 + \ldots\ldots$ upto 10 terms is:

(A) 11300 (B) 11200 (C) 12100 (D) 12300

Open solution

**Q297.** JEE Main 2013 (Online) 22th April Morning Slot · MCQ (Single Correct Answer)

The sum $\frac{3}{1^2} + \frac{5}{1^2+2^2} + \frac{7}{1^2+2^2+3^2} + \ldots\ldots$ upto 11 terms is

(A) $\frac{7}{2}$ (B) $\frac{11}{4}$ (C) $\frac{11}{2}$ (D) $\frac{60}{11}$

Open solution

**Q298.** JEE Main 2013 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The sum of the series $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \ldots\ldots$ upto 10 terms is

(A) $\frac{18}{11}$ (B) $\frac{22}{13}$ (C) $\frac{20}{11}$ (D) $\frac{16}{9}$

Open solution

**Q299.** JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

The sum of first 20 terms of the sequence 0.7, 0.77, 0.777,.....,is

(A) $\frac{7}{81}\left(179 - 10^{-20}\right)$ (B) $\frac{7}{9}\left(99 - 10^{-20}\right)$
(C) $\frac{7}{81}\left(179 + 10^{-20}\right)$ (D) $\frac{7}{9}\left(99 + 10^{-20}\right)$

Open solution

**Q300.** AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

If the sum of the series $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \ldots$ upto $n$ terms, when $n$ is even, is $\frac{n(n+1)^2}{2}$, then the sum of the series, when $n$ is odd, is

(A) $n^2(n+1)$ (B) $\frac{n^2(n-1)}{2}$ (C) $\frac{n^2(n+1)}{2}$ (D) $n^2(n-1)$

Open solution

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**Q301.** AIEEE 2012 (Online) 19th May · MCQ (Single Correct Answer)

The sum of the series $1+\frac{4}{3}+\frac{10}{9}+\frac{28}{27}+\ldots$ upto $n$ terms is

(A) $\frac{7}{6}n+\frac{1}{6}-\frac{2}{3\cdot 2^{n-1}}$ (B) $\frac{5}{3}n-\frac{7}{6}+\frac{1}{2\cdot 3^{n-1}}$

(C) $n+\frac{1}{2}-\frac{1}{2\cdot 3^{n}}$ (D) $n-\frac{1}{3}-\frac{1}{3\cdot 2^{n-1}}$

Open solution

**Q302.** AIEEE 2012 (Online) 12th May · MCQ (Single Correct Answer)

The sum of the series
$\frac{1}{1+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+\ldots$
upto 15 terms is

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

**Q303.** AIEEE 2012 (Online) 7th May · MCQ (Single Correct Answer)

The sum of the series $1^2+2\cdot 2^2+3^2+2\cdot 4^2+5^2+2\cdot 6^2+\ldots\ldots+2(2m)^2$ is

(A) $m(2m+1)^2$ (B) $m^2(m+2)$ (C) $m^2(2m+1)$ (D) $m(m+2)^2$

Open solution

**Q304.** AIEEE 2012 · MCQ (Single Correct Answer)

**Statement-1:** The sum of the series 1 + (1 + 2 + 4) + (4 + 6 + 9) + (9 + 12 + 16) +.....+ (361 + 380 + 400) is 8000.

**Statement-2:** $\sum_{k=1}^{n}\left(k^3-(k-1)^3\right)=n^3$, for any natural number n.

(A) Statement-1 is false, Statement-2 is true.

(B) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

(C) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

(D) Statement-1 is true, Statement-2 is false.

Open solution

**Q305.** AIEEE 2009 · MCQ (Single Correct Answer)

The sum to infinite term of the series $1+\frac{2}{3}+\frac{6}{3^2}+\frac{10}{3^3}+\frac{14}{3^4}+\ldots\ldots$ is

(A) 3 (B) 4 (C) 6 (D) 2

Open solution

**Q306.** AIEEE 2007 · MCQ (Single Correct Answer)

The sum of series $\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\ldots\ldots$ upto infinity is

(A) $e^{-\frac{1}{2}}$ (B) $e^{+\frac{1}{2}}$ (C) $e^{-2}$ (D) $e^{-1}$

Open solution

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**Q307.** AIEEE 2005 · MCQ (Single Correct Answer)

The sum of the series $1+\frac{1}{4.2!}+\frac{1}{16.4!}+\frac{1}{64.6!}+\ldots\ldots$ ad inf. is

(A) $\frac{e-1}{\sqrt{e}}$ (B) $\frac{e+1}{\sqrt{e}}$ (C) $\frac{e-1}{2\sqrt{e}}$ (D) $\frac{e+1}{2\sqrt{e}}$

Open solution

**Q308.** AIEEE 2004 · MCQ (Single Correct Answer)

The sum of the first n terms of the series $1^2+2.2^2+3^2+2.4^2+5^2+2.6^2+\ldots.$ $is$ $\frac{n(n+1)^2}{2}$ when n is even. When n is odd the sum is

(A) $\left[\frac{n(n+1)}{2}\right]^2$ (B) $\frac{n^2(n+1)}{2}$

(C) $\frac{n(n+1)^2}{4}$ (D) $\frac{3n(n+1)}{2}$

Open solution

**Q309.** AIEEE 2004 · MCQ (Single Correct Answer)

The sum of series $\frac{1}{2!}+\frac{1}{4!}+\frac{1}{6!}+\ldots\ldots\ldots$ is

(A) $\frac{(e^2-2)}{e}$ (B) $\frac{(e-1)^2}{2e}$

(C) $\frac{(e^2-1)}{2e}$ (D) $\frac{(e^2-1)}{2}$

Open solution

**Q310.** AIEEE 2003 · MCQ (Single Correct Answer)

The sum of the series $\frac{1}{1.2}-\frac{1}{2.3}+\frac{1}{3.4}\ldots\ldots\ldots\ldots\ldots$ up to $\infty$ is equal to

(A) $\log_e\left(\frac{4}{e}\right)$ (B) $2\log_e 2$

(C) $\log_e 2-1$ (D) $\log_e 2$

Open solution

**Q311.** AIEEE 2002 · MCQ (Single Correct Answer)

The value of $2^{1/4}.\ 4^{1/8}.\ 8^{1/16}\ldots\infty$ is

(A) 1 (B) 2 (C) 3/2 (D) 4

Open solution

**Q312.** AIEEE 2002 · MCQ (Single Correct Answer)

$1^3-2^3+3^3-4^3+\ldots+9^3=$

(A) 425 (B) - 425 (C) 475 (D) - 475

Open solution

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**Q313.** JEE Main 2023 (Online) 15th April Morning Shift · Numerical

If the sum of the series

$$\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{2^2}-\frac{1}{2\cdot 3}+\frac{1}{3^2}\right)+\left(\frac{1}{2^3}-\frac{1}{2^2\cdot 3}+\frac{1}{2\cdot 3^2}-\frac{1}{3^3}\right)+$$

$$\left(\frac{1}{2^4}-\frac{1}{2^3\cdot 3}+\frac{1}{2^2\cdot 3^2}-\frac{1}{2\cdot 3^3}+\frac{1}{3^4}\right)+\ldots$$

is $\frac{\alpha}{\beta}$, where $\alpha$ and $\beta$ are co-prime, then $\alpha+3\beta$ is equal to _____.

Open solution

**Q314.** JEE Main 2023 (Online) 13th April Morning Shift · Numerical

The sum to 20 terms of the series $2\cdot 2^2-3^2+2\cdot 4^2-5^2+2\cdot 6^2-\ldots\ldots$ is equal to _____.

Open solution

**Q315.** JEE Main 2023 (Online) 11th April Evening Shift · Numerical

For $k\in\mathbb{N}$, if the sum of the series $1+\frac{4}{k}+\frac{8}{k^2}+\frac{13}{k^3}+\frac{19}{k^4}+\ldots$ is 10 , then the value of $k$ is _____.

Open solution

**Q316.** JEE Main 2023 (Online) 11th April Morning Shift · Numerical

Let $S=109+\frac{108}{5}+\frac{107}{5^2}+\ldots.+\frac{2}{5^{107}}+\frac{1}{5^{108}}$. Then the value of $\left(16S-(25)^{-54}\right)$ is equal to _____.

Open solution

**Q317.** JEE Main 2023 (Online) 10th April Evening Shift · Numerical

Suppose $a_1, a_2, 2, a_3, a_4$ be in an arithmetico-geometric progression. If the common ratio of the corresponding geometric progression is 2 and the sum of all 5 terms of the arithmetico-geometric progression is $\frac{49}{2}$, then $a_4$ is equal to _____.

Open solution

**Q318.** JEE Main 2023 (Online) 10th April Morning Shift · Numerical

The sum of all those terms, of the arithmetic progression 3, 8, 13, ...., 373, which are not divisible by 3, is equal to _____.

Open solution

**Q319.** JEE Main 2023 (Online) 8th April Evening Shift · Numerical

Let $0<z<y<x$ be three real numbers such that $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in an arithmetic progression and $x, \sqrt{2}y, z$ are in a geometric progression. If $xy+yz+zx=\frac{3}{\sqrt{2}}xyz$, then $3(x+y+z)^2$ is equal to _____.

Open solution

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**Q320.** JEE Main 2023 (Online) 31st January Evening Shift · Numerical

The sum $1^2 - 2 \cdot 3^2 + 3 \cdot 5^2 - 4 \cdot 7^2 + 5 \cdot 9^2 - \ldots + 15 \cdot 29^2$ is _____.

Open solution

**Q321.** JEE Main 2023 (Online) 30th January Evening Shift · Numerical

The $8^{th}$ common term of the series

$S_1 = 3 + 7 + 11 + 15 + 19 + \ldots\ldots$

$S_2 = 1 + 6 + 11 + 16 + 21 + \ldots\ldots$

is :

Open solution

**Q322.** JEE Main 2023 (Online) 30th January Morning Shift · Numerical

Let $\sum_{n=0}^{\infty} \frac{n^3((2n)!)+(2n-1)(n!)}{(n!)((2n)!)} = ae + \frac{b}{e} + c$, where $a, b, c \in \mathbb{Z}$ and $e = \sum_{n=0}^{\infty} \frac{1}{n!}$ Then $a^2 - b + c$ is equal to _____.

Open solution

**Q323.** JEE Main 2023 (Online) 29th January Evening Shift · Numerical

Let $a_1 = b_1 = 1$ and $a_n = a_{n-1} + (n-1), b_n = b_{n-1} + a_{n-1}, \forall n \geq 2$. If $S = \sum_{n=1}^{10} \frac{b_n}{2^n}$ and $T = \sum_{n=1}^{8} \frac{n}{2^{n-1}}$, then $2^7(2S - T)$ is equal to _____.

Open solution

**Q324.** JEE Main 2023 (Online) 24th January Evening Shift · Numerical

If $\frac{1^3+2^3+3^3+\ldots \text{up to } n \text{ terms}}{1.3+2.5+3.7+\ldots \text{up to } n \text{ terms}} = \frac{9}{5}$, then the value of $n$ is

Open solution

**Q325.** JEE Main 2022 (Online) 29th July Morning Shift · Numerical

If $\frac{1}{2\times3\times4} + \frac{1}{3\times4\times5} + \frac{1}{4\times5\times6} + \ldots + \frac{1}{100\times101\times102} = \frac{k}{101}$, then 34 k is equal to _____.

Open solution

**Q326.** JEE Main 2022 (Online) 28th July Evening Shift · Numerical

If $\frac{6}{3^{12}} + \frac{10}{3^{11}} + \frac{20}{3^{10}} + \frac{40}{3^9} + \ldots + \frac{10240}{3} = 2^n \cdot m$, where m is odd, then m . n is equal to _____.

Open solution

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**Q327.** JEE Main 2022 (Online) 27th July Evening Shift · Numerical

$\frac{2^3-1^3}{1\times7}+\frac{4^3-3^3+2^3-1^3}{2\times11}+\frac{6^3-5^3+4^3-3^3+2^3-1^3}{3\times15}+\cdots+\frac{30^3-29^3+28^3-27^3+\ldots+2^3-1^3}{15\times63}$ is equal to _____.

Open solution

**Q328.** JEE Main 2022 (Online) 26th July Evening Shift · Numerical

If $\sum_{k=1}^{10}\frac{k}{k^4+k^2+1}=\frac{m}{n}$, where m and n are co-prime, then $m+n$ is equal to _____.

Open solution

**Q329.** JEE Main 2022 (Online) 26th July Morning Shift · Numerical

The series of positive multiples of 3 is divided into sets : $\{3\},\{6,9,12\},\{15,18,21,24,27\},\ldots$ Then the sum of the elements in the $11^{th}$ set is equal to _____.

Open solution

**Q330.** JEE Main 2022 (Online) 25th July Morning Shift · Numerical

Let $a_1=b_1=1, a_n=a_{n-1}+2$ and $b_n=a_n+b_{n-1}$ for every

natural number $n\geqslant 2$. Then $\sum_{n=1}^{15}a_n.b_n$ is equal to _____.

Open solution

**Q331.** JEE Main 2022 (Online) 30th June Morning Shift · Numerical

Let for $f(x)=a_0x^2+a_1x+a_2,\ f'(0)=1$ and $f'(1)=0$. If $a_0$, $a_1$, $a_2$ are in an arithmatico-geometric progression, whose corresponding A.P. has common difference 1 and corresponding G.P. has common ratio 2, then f(4) is equal to _____.

Open solution

**Q332.** JEE Main 2022 (Online) 27th June Morning Shift · Numerical

If the sum of the first ten terms of the series

$\frac{1}{5}+\frac{2}{65}+\frac{3}{325}+\frac{4}{1025}+\frac{5}{2501}+\ldots.$

is $\frac{m}{n}$, where m and n are co-prime numbers, then m + n is equal to _____.

Open solution

**Q333.** JEE Main 2022 (Online) 26th June Morning Shift · Numerical

Let $A=\sum_{i=1}^{10}\sum_{j=1}^{10}\min\{i,j\}$ and $B=\sum_{i=1}^{10}\sum_{j=1}^{10}\max\{i,j\}$. Then A + B is equal to _____.

Open solution

**Q334.** JEE Main 2022 (Online) 25th June Morning Shift · Numerical

The greatest integer less than or equal to the sum of first 100 terms of the sequence $\frac{1}{3},\frac{5}{9},\frac{19}{27},\frac{65}{81},$ ..... is equal to _____.

Open solution

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**Q335.** JEE Main 2021 (Online) 31st August Evening Shift · **Numerical**

If $S = \frac{7}{5} + \frac{9}{5^2} + \frac{13}{5^3} + \frac{19}{5^4} + \ldots$, then 160 S is equal to _____.

Open solution

**Q336.** JEE Main 2021 (Online) 25th July Morning Shift · **Numerical**

If the value of

$$\left(1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \ldots . upto\ \infty\right)^{\log_{(0.25)}\left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \ldots . upto\ \infty\right)}$$

is $l$, then $l^2$ is equal to _____.

Open solution

**Q337.** JEE Main 2021 (Online) 20th July Evening Shift · **Numerical**

For k $\in$ N, let $\frac{1}{\alpha(\alpha+1)(\alpha+2)\ldots\ldots\ldots(\alpha+20)} = \sum_{k=0}^{20} \frac{A_k}{\alpha+k}$, where $\alpha > 0$. Then the value of $100\left(\frac{A_{14}+A_{15}}{A_{13}}\right)^2$ is equal to _____.

Open solution

**Q338.** JEE Main 2021 (Online) 20th July Evening Shift · **Numerical**

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_1 = 1$, $a_2 = 1$ and $a_{n+2} = 2a_{n+1} + a_n$ for all $n \geq 1$. Then the value of $47 \sum_{n=1}^{\infty} \frac{a_n}{2^{3n}}$ is equal to _____.

Open solution

**Q339.** JEE Main 2021 (Online) 16th March Evening Shift · **Numerical**

$S_n(x) = \log_{a^{1/2}}x + \log_{a^{1/3}}x + \log_{a^{1/6}}x + \log_{a^{1/11}}x + \log_{a^{1/18}}x + \log_{a^{1/27}}x + \ldots\ldots$ up to n-terms, where a > 1. If $S_{24}(x) = 1093$ and $S_{12}(2x) = 265$, then value of a is equal to _____.

Open solution

**Q340.** JEE Main 2020 (Online) 8th January Evening Slot · **Numerical**

The sum, $\sum_{n=1}^{7} \frac{n(n+1)(2n+1)}{4}$ is equal to _____.

Open solution

**Q341.** JEE Main 2020 (Online) 8th January Morning Slot · **Numerical**

The sum $\sum_{k=1}^{20} (1 + 2 + 3 + \ldots + k)$ is :

Open solution

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# Answer Key

## MCQ (Single Correct Answer)

| | | | | |
|---|---|---|---|---|
| 1. **A** | 2. **C** | 3. **C** | 4. **B** | 5. **A** |
| 6. **C** | 7. **C** | 8. **D** | 9. **C** | 10. **B** |
| 11. **A** | 12. **D** | 13. **B** | 14. **B** | 15. **A** |
| 16. **A** | 17. **D** | 18. **B** | 19. **A** | 20. **D** |
| 21. **A** | 22. **A** | 23. **B** | 24. **D** | 25. **A** |
| 26. **B** | 27. **B** | 28. **D** | 29. **D** | 30. **A** |
| 31. **B** | 32. **B** | 33. **A** | 34. **D** | 35. **B** |
| 36. **B** | 37. **B** | 38. **A** | 39. **B** | 40. **C** |
| 41. **B** | 42. **B** | 43. **B** | 44. **B** | 45. **D** |
| 46. **D** | 47. **D** | 48. **C** | 49. **D** | 50. **B** |
| 51. **C** | 52. **A** | 53. **C** | 54. **B** | 55. **C** |
| 56. **A** | 57. **B** | 58. **A** | 59. **A** | 60. **C** |
| 61. **C** | 62. **D** | 63. **D** | 64. **B** | 65. **B** |
| 66. **C** | 67. **D** | 68. **B** | 69. **A** | 70. **B** |
| 71. **D** | 72. **A** | 73. **D** | 74. **A** | 75. **D** |
| 76. **C** | 77. **C** | 78. **B** | 79. **B** | 80. **A** |
| 81. **A** | 82. **B** | 83. **D** | 84. **C** | 85. **A** |
| 86. **B** | 87. **C** | 88. **C** | 89. **C** | 90. **D** |
| 91. **C** | 92. **D** | 93. **D** | 94. **B** | 95. **B** |
| 96. **C** | 97. **B** | 98. **B** | 99. **A** | 100. **A** |
| 101. **C** | 102. **D** | 103. **D** | 104. **D** | 105. **B** |
| 106. **D** | 107. **A** | 108. **B** | 109. **C** | 110. **D** |
| 111. **D** | 112. **D** | 113. **A** | 114. **B** | 115. **C** |
| 116. **B** | 117. **D** | 118. **B** | 119. **C** | 120. **A** |
| 121. **A** | 122. **C** | 123. **C** | 124. **A** | 125. **C** |
| 126. **D** | 127. **B** | 128. **A** | 129. **D** | 130. **B** |
| 131. **D** | 132. **B** | 133. **C** | 134. **D** | 135. **D** |
| 136. **A** | 137. **D** | 138. **A** | 139. **C** | 140. **D** |
| 141. **D** | 142. **C** | 143. **A** | 144. **C** | 145. **A** |
| 146. **D** | 147. **A** | 148. **C** | 149. **D** | 150. **D** |

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| | | | | |
|---|---|---|---|---|
| 151. **B** | 152. **C** | 153. **D** | 154. **B** | 155. **A** |
| 156. **C** | 157. **C** | 158. **B** | 159. **C** | 160. **B** |
| 161. **B** | 162. **C** | 163. **B** | 164. *Scan QR* | 165. **D** |
| 166. **D** | 167. **C** | 168. **D** | 169. **C** | 170. **A** |
| 171. **B** | 172. **B** | 173. **D** | 174. **D** | 175. **D** |
| 176. **D** | 177. **B** | 178. **B** | 179. **D** | 180. **B** |
| 181. **B** | | | | |

## Numerical

| | | | | |
|---|---|---|---|---|
| 182. **1279** | 183. **91** | 184. **976** | 185. **90** | 186. **9** |
| 187. **2** | 188. **441** | 189. **11132** | 190. **20** | 191. **474** |
| 192. **1011** | 193. **1505** | 194. **103** | 195. **3660** | 196. **6** |
| 197. **76** | 198. **910** | 199. **1** | 200. **6699** | 201. **9** |
| 202. **353** | 203. **9** | 204. **400** | 205. **321** | 206. **754** |
| 207. **8** | 208. **9** | 209. **60** | 210. **3** | 211. **12** |
| 212. **16** | 213. **53** | 214. **38** | 215. **2223** | 216. **41651** |
| 217. **702** | 218. **40** | 219. **7744** | 220. **2021** | 221. **3** |
| 222. **1251** | 223. **14** | 224. **3** | 225. **10** | 226. **9** |
| 227. **3** | 228. **39** | 229. **4** | 230. **14** | |

## Out of syllabus

| | | | | |
|---|---|---|---|---|
| 231. **C** | 232. **D** | 233. **B** | 234. **D** | 235. **C** |
| 236. **D** | 237. **B** | 238. **C** | 239. **C** | 240. *Scan QR* |
| 241. **B** | 242. **B** | 243. **C** | 244. **C** | 245. **B** |
| 246. **B** | 247. **C** | 248. **B** | 249. **C** | 250. **C** |
| 251. **B** | 252. **A** | 253. **B** | 254. **A** | 255. **A** |
| 256. **D** | 257. **D** | 258. **D** | 259. **B** | 260. **B** |
| 261. **B** | 262. **B** | 263. **C** | 264. **C** | 265. **D** |
| 266. **A** | 267. **C** | 268. **A** | 269. **B** | 270. **B** |
| 271. **A** | 272. **C** | 273. **A** | 274. **B** | 275. **B** |
| 276. **D** | 277. **A** | 278. **C** | 279. **D** | 280. **D** |
| 281. **A** | 282. **C** | 283. **B** | 284. **A** | 285. **B** |
| 286. **B** | 287. **D** | 288. **B** | 289. **D** | 290. **A** |
| 291. **B** | 292. **D** | 293. **A** | 294. **A** | 295. **A** |

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| | | | | |
|---|---|---|---|---|
| 296. **C** | 297. **C** | 298. **C** | 299. **C** | 300. **C** |
| 301. *Scan QR* | 302. **C** | 303. **A** | 304. **B** | 305. **A** |
| 306. **D** | 307. **D** | 308. **B** | 309. **B** | 310. **A** |
| 311. **B** | 312. **A** | 313. **7** | 314. **1310** | 315. **2** |
| 316. **2175** | 317. **16** | 318. **9525** | 319. **150** | 320. **6952** |
| 321. **151** | 322. **26** | 323. **461** | 324. **5** | 325. **286** |
| 326. **12** | 327. **120** | 328. **166** | 329. **6993** | 330. **27560** |
| 331. **2** | 332. **276** | 333. **1100** | 334. **98** | 335. **305** |
| 336. **3** | 337. **9** | 338. **7** | 339. **16** | 340. **504** |
| 341. **1540** | | | | |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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